

Chapter 9

Mechanical Design

Structural design, load analysis, material selection, fabrication, and validation of the robotic arm.

9.1 Introduction

In this chapter we walk through the mechanical design of our robotic arm and how we validated it. The arm is what physically does the work in our system: it carries the end-effector across the workspace, holds it still while the camera identifies objects, and moves the gripped ball over to the right container. Every other part of the project — perception, control, integration — only matters if the arm itself doesn't break under load. That's why we treated the mechanical side as the foundation everything else sits on.

We structured the chapter as a funnel. We start with the geometry (the link lengths and joint layout), then use that to work out the joint torques and verify the inherited motors. From there we describe the beam material, design the cross-sections, run the static analysis and FEM, and finally cover the assembly details, the base rotation system, and the as-built mass. Sections on testing, requirements, limitations, and future work close the chapter.

The methods are the ones from our mechanical engineering coursework: Grübler's mobility equation for counting degrees of freedom, cantilever beam theory for the statics, the von Mises criterion for combined stress in the FEM, and the manufacturer datasheets for the materials. We address mechanical requirements M1 through M4 from the master requirements list, and we set ourselves a self-imposed target of $n \geq 2.0$ for the Factor of Safety on the structural beams. We picked 2.0 because FDM-printed parts come with real uncertainty — the layers don't always bond perfectly, the material isn't fully isotropic, and dimensional tolerances vary from print to print. Mott (2014) recommends a similar margin for components in this kind of uncertainty bracket, which lined up with our judgement.

Where this chapter departs from the previous group's report (Gruppe 12, Sortify): we inherited the serial-revolute architecture and the Dynamixel motors, but we redesigned every structural part. Their carbon-fibre beams broke during our initial testing, so we redesigned in PLA. We changed the base rotation assembly so the rotation motor only rotates and doesn't carry weight. We changed the gripper from a small servo to an XM-430 with current-controlled gripping. We documented what was carried over and what was replaced as honestly as possible throughout the chapter.



Figure 9.1 — The as-built arm in working configuration on its test rig, mid-task with the ball pickup tray.

9.2 Design Requirements

The mechanical design has to satisfy the project's master requirements M1 through M4. M1 fixes the workspace the arm must reach. M2 sets the mass limit of objects the arm must handle. M3 covers operating speed and cycle time at a system level. M4 sets the durability target — the arm must complete the demonstration runs without structural failure. Of these, M1 and M4 are squarely mechanical; M2 and M3 are mostly integration-level concerns that involve mechanical, electrical, and control together. Verification against these requirements is gathered in §9.14.

9.3 Concept Choice

The arm architecture is a serial revolute manipulator: five rotational joints (base rotation, shoulder, elbow-1, elbow-2, gripper drive) connected by three structural beams (long, middle, short) plus the end-effector and the base rotation platform. We inherited this architecture from the previous group along with the Dynamixel motors. Our concept work therefore focused on the iterations within that architecture rather than on choosing a manipulator topology — we did not run a trade study between serial-revolute, SCARA, or Cartesian options.

What we did decide at the concept level: we kept the inherited architecture, we kept the inherited motor set, and we redesigned every other part. The previous group's beams were carbon-fibre, but they had broken before we received the arm and they broke again during our early testing. We replaced them with PLA printed on the in-house Bambu Lab P1S printer. The base rotation system was reworked so the base motor handles only rotation, with a separate bearing ring carrying the vertical load. The end-effector was changed from a small servo gripper to an XM-430 with current-controlled gripping (giving us position feedback and a soft grip without needing extra sensors). Each of those decisions is justified in detail in the section it belongs to.

9.4 Geometry, Degrees of Freedom, and Reach

9.4.1 Link and joint inventory

The arm has five motors above the base, three structural beams between them, and an end-effector at the tip. Working from the base outward:

- Base rotation joint: XM-430 mounted in the lid, rotates the entire arm 360° about the vertical axis.
- Long beam: 220.52 mm long, 44 × 40.5 mm cross-section, 5 mm walls plus 2 mm internal triangular ribs, 219.38 g in the CAD model at 100 % infill (lighter as printed at 40–60 % gyroid infill).
- Shoulder joint: XM-540 driven, mounted in a custom bracket inside the lid, rotates the arm in the vertical plane.
- Middle beam: 188 mm long, ~36 × 30 mm cross-section, 2.5 mm walls plus internal ribs, 110.88 g in CAD.
- First elbow joint: XM-430.
- Short beam: 114.55 mm long, 36.42 × ~30 mm cross-section, 2 mm walls plus internal ribs, 62.11 g in CAD. Carries the camera mount and the gripper bracket.
- Second elbow joint: XM-430.
- Gripper drive: XM-430-W210 with a two-finger four-contact-point claw, current-limited grip.
- Camera: 91 g, mounted on the short beam via support brackets and an L-bracket with slotted angle adjustment.

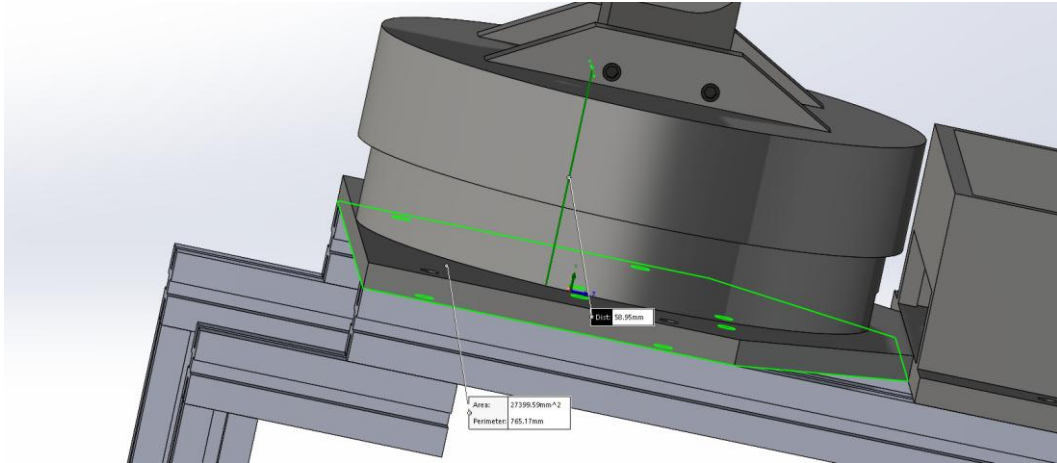


Figure 9.2 — Vertical stack-up from the rotation platform up to the XM-540 mounting face: 58.95 mm.

9.4.2 Degree-of-freedom analysis

Grübler's mobility equation for a spatial open chain: $M = 6(n - 1) - \sum(6 - f_i)$, where n is the number of links including the ground, and f_i is the number of degrees of freedom at joint i . For our arm: $n = 6$ (ground + 5 movable links), 5 revolute joints (each $f_i = 1$).

$$M = 6(6 - 1) - (5 \times 5) = 30 - 25 = 5$$

Five active degrees of freedom. The end-effector pose in 3D requires six DoF (three translations, three rotations), so the arm is under-actuated relative to general 6-DoF tasks. However, for the project task (pick a ball from one of a finite set of positions and place it in one of a finite set of containers), five DoF is sufficient: the base rotation provides azimuth, the shoulder and two elbows provide reach and height, and the gripper drive opens and closes. The end-effector pitch is set by the joint configuration; explicit roll is not required.

9.4.3 Reach

We measured reach in the fully extended horizontal pose, which is the worst case for bending at every joint and the case we use for sizing the cross-sections. Two reach numbers matter: the centre-to-centre distance from the shoulder motor (XM-540) to the tip of the short beam (before the claw), and the total reach to the outermost point of the static finger.

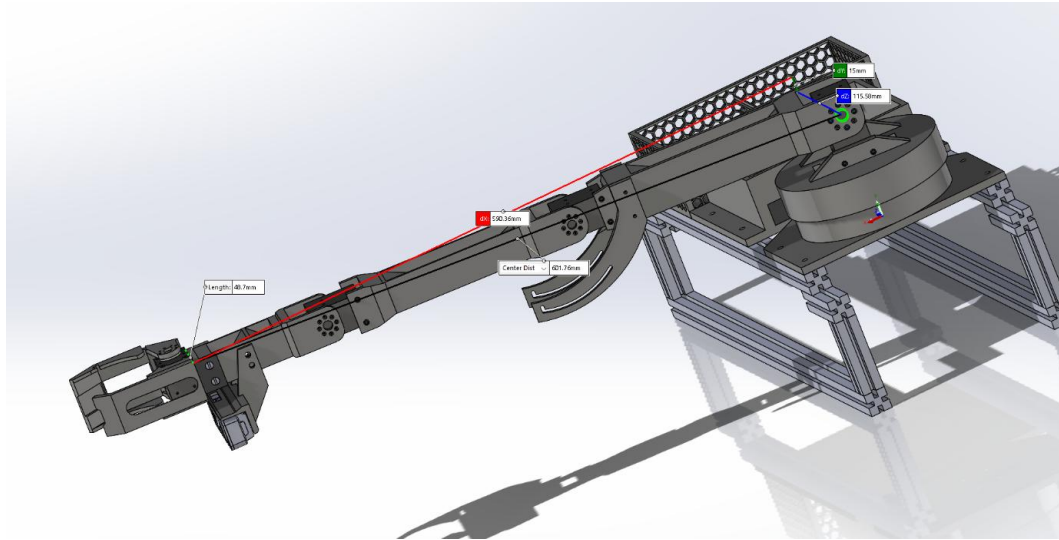


Figure 9.3 — Arm in the fully extended horizontal pose, isometric view. Centre distance from the shoulder motor to the short-beam tip: 601.76 mm.

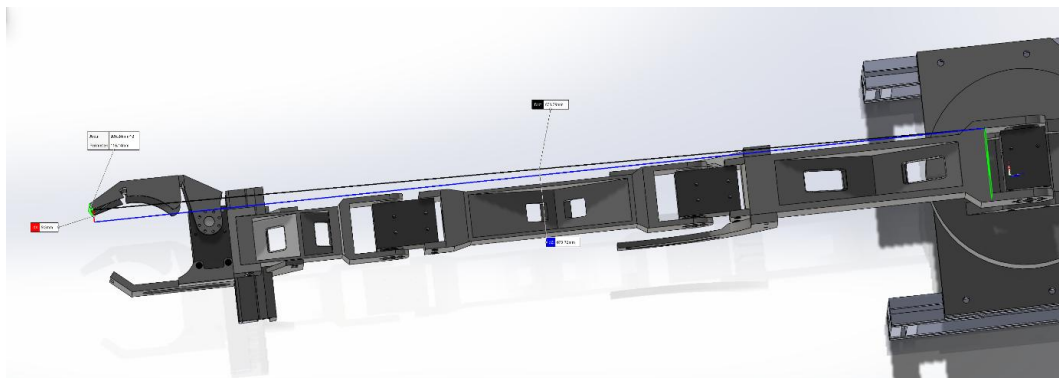


Figure 9.4 — Same pose with the gripper attached. Total reach from the shoulder centre to the static-finger tip: 671.79 mm.

601.76 mm to the short-beam tip; 671.79 mm to the claw tip. The additional 70 mm is the offset from the short-beam tip out through the gripper assembly to the static-finger tip. That's a meaningful increase over the previous group's arm, which fell short of the platform edge in our test setup. The extra length came from re-dimensioning the three beams to give clearance above the platform plus enough reach to pick objects off the floor in front of the platform.

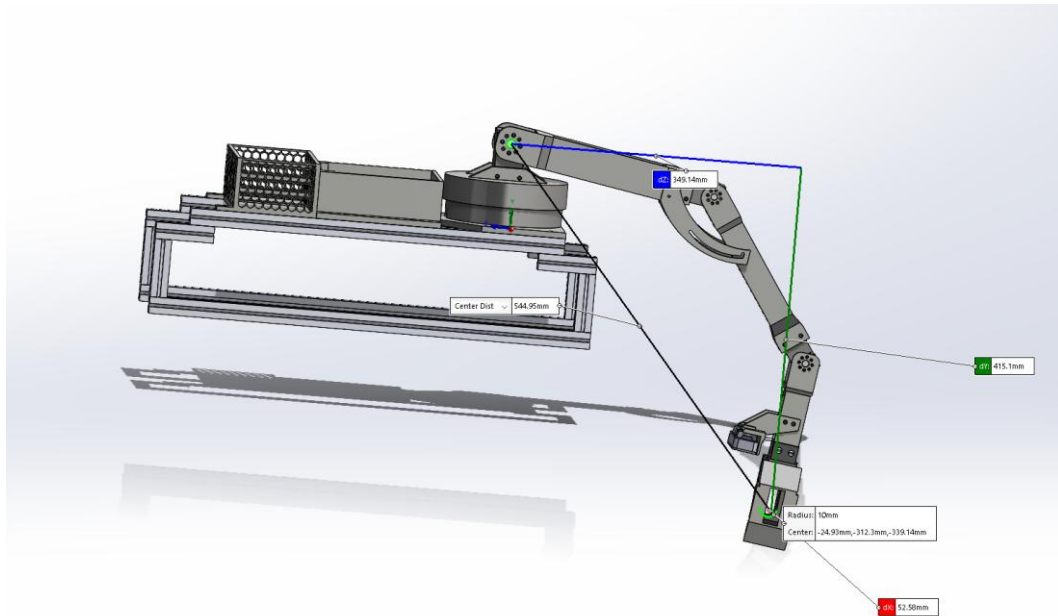


Figure 9.5 — Worst-case low-reach pose. Vertical drop from shoulder centre to gripper tip is 415.1 mm; straight-line distance from shoulder to gripper is 544.95 mm; horizontal projection 349.14 mm.

The fully extended horizontal pose with reach 671.79 mm to the claw tip is the worst-case load configuration and is the configuration used throughout the rest of the chapter for the structural calculations. The arm exceeds the workspace required by M1 with margin to spare.

9.5 Actuator and Load Analysis

9.5.1 Method

The arm carries the end-effector and any payload as a chain of masses cantilevered from the base. To verify the inherited Dynamixel motors are adequate for the cascade and to size the structural beams, we built a downstream-mass cascade from the end-effector inward. At each joint we compute the cumulative mass that joint has to lift in the worst-case fully extended horizontal pose, then multiply by the moment arm to get the torque the joint motor sees. We compare against the motor's rated continuous torque, not stall, since the arm operates continuously during pick-and-place tasks and we want to stay within the motor's safe long-term envelope.

Sign and coordinate convention: positive vertical is upward, and moment arms are measured as the perpendicular distance from the joint axis to each point load in the worst-case horizontal pose. Tip loads on each beam are taken as point loads at the distal end; self-weight is taken as a point load at the beam midspan.

The end-effector subsystem and its contents are at the short beam's tip. Component masses, measured from the CAD model (which matches the printed parts to within $\pm 5\%$ for the structural plastic parts):

- Camera: 91 g
- Camera mount (L-bracket): 14 g

- Camera support brackets (2 × 5 g): 10 g
- Claw mount bracket: 13.52 g
- Static jaw: 14 g
- Movable jaw: 25 g
- Gripper-drive XM-430-W210 motor: 82 g
- Payload (target object, e.g. ball): 200 g
- End-effector subtotal (everything past the short-beam tip): 449.5 g, weight 4.41 N

As we walk base-ward, we add each beam's self-weight and the joint motor mounted at that joint. Beam self-weights are taken at the CAD value (100 % infill); printed parts at 40–60 % gyroid infill weigh less, so using CAD mass is conservative for the load cascade.

9.5.2 Short beam (wrist joint)

Tip load on the short beam is the end-effector subtotal: 4.41 N. The short beam itself adds 62.11 g (0.61 N) of distributed self-weight. The second-elbow XM-430 physically sits between the short and middle beams, but for the mass cascade we account its weight to the middle-beam side of the joint: from the short beam's perspective, that motor is at the proximal pivot (not the distal tip), so it does not contribute to the short beam's tip load. Its weight enters the cascade at the next step up (§9.5.3, middle beam).

$$F_{\text{tip,short}} = 4.41 \text{ N}$$

9.5.3 Middle beam (elbow joint)

Tip load on the middle beam = short beam end-effector + short beam self-weight + second-elbow motor that lives at the middle-beam tip:

$$F_{\text{tip,middle}} = 4.41 + 0.61 + 0.80 = 5.82 \text{ N}$$

The middle beam itself adds 110.88 g (1.09 N) of distributed self-weight.

9.5.4 Long beam (shoulder joint)

Tip load on the long beam = everything past the first elbow joint + first-elbow motor:

$$F_{\text{tip,long}} = 5.82 + 1.09 + 0.80 = 7.71 \text{ N}$$

The long beam itself adds 219.38 g (2.15 N) of distributed self-weight.

9.5.5 Cascade summary

The three beam tip loads in the worst-case horizontal extended pose are 4.41 N (short), 5.82 N (middle), and 7.71 N (long). At the shoulder pivot, the total reaction force is approximately 9.86 N (long beam tip load + long beam self-weight), and the moment about the shoulder pivot is approximately 1.94 N·m. The shoulder XM-540 has a rated continuous torque well above this figure, with substantial margin for dynamic moves and acceleration. The two XM-430 elbow motors face moments of approximately 1.20 N·m and 0.54 N·m at the first and second elbow respectively, again well below their continuous-rated torque.

All four joint motors have sufficient torque headroom against the cascaded static loads. The tip loads 4.41 / 5.82 / 7.71 N at the short, middle, and long beams are the loads used as inputs to the structural analyses in §9.8 and §9.10.

9.6 Material Selection

9.6.1 Why PLA

We picked PLA for all the structural beams and brackets. The reasons were practical: USN's mechanical lab has a Bambu Lab P1S in working order, with Bambu PLA Basic filament in stock at no additional cost to the project, and we already had experience using both. That meant we could iterate the design in hours rather than days — important when we found early that the inherited carbon-fibre beams were broken and needed a full redesign. PLA prints cleanly without needing an enclosed-chamber printer or special filament handling, and at the loads we're seeing (sub-10 N tip loads, sub-2 N·m joint moments) its strength is more than adequate. Detailed strength margins are computed in §9.8 and §9.10.

We also considered Nylon 12 (Formlabs SLS process). Nylon 12 has real engineering advantages over FDM PLA: it's effectively isotropic because the SLS process bonds powder in all three directions, it's ductile rather than brittle (so it would yield before failing in an overload), and it would not have the layer-bonding weakness that FDM PLA has at the bracket-to-beam interfaces. But Nylon 12 prints would have meant outsourcing to a service bureau with a turnaround we couldn't fit into the project schedule. We ran a preliminary FEM with Nylon 12 properties to characterise what we would have lost in stiffness ($E \approx 1.9$ GPa for Nylon 12 versus 3.5 GPa for PLA — a 45 % lower Young's modulus, meaning deflection would roughly double for the same geometry). The detail and result of that study is in §9.18 as recommended future work.

9.6.2 Print recipe

The early prints failed at the bracket-to-beam interfaces and at the gusset junctions — the cracks were unambiguously along the layer lines. That told us layer adhesion was the issue, not bulk material strength. We developed a print recipe that targets stronger layer bonding while keeping print time manageable:

- Wall loops: 6 (we had also tried 8 on the long beam; 6 was the practical compromise of strength versus print time)
- Sparse infill density: 40 % (some long-beam prints went to 60 % for the highest-loaded sections)
- Sparse infill pattern: gyroid (better isotropy and shear performance than rectilinear)
- Outer wall speed: 35 mm/s (slower than default; gives the polymer more time to bond to the layer below)
- Max fan speed: 60 % (reduced from default 100 % — cooler walls bond less well, so we keep them slightly warmer)
- Nozzle temperature: 225–230 °C (top of the Bambu PLA Basic recommended range)

- Filament: Bambu PLA Basic, 0.4 mm nozzle, P1S printer

Filament settings

* Bambu PLA Basic @BBL P1S 0.4 nozzle

Advanced

Filament

Cooling

Setting Overrides

Advanced

Notes

Multi Filament

Direct Drive: Standard

Direct Drive: High Flow

Wiki

Velocity Adaptation Factor

1

Price

24.99 money/kg

Softening temperature

45

Filament prime volume

Filament change

30 mm³

Hotend change

60 mm³

Filament ramming length

Extruder change

5 mm

Hotend change

10 mm

Travel time after ramming

Extruder change

0 s

Hotend change

0 s

Precooling target temperature

Extruder change

0 °C

Hotend change

0 °C

Recommended nozzle temperature

Min

225 °C

Max

230 °C

Print temperature

Cool Plate SuperTack

Initial layer

45 °C

Other layers

45 °C

Cool Plate

Initial layer

35 °C

Other layers

35 °C

Engineering Plate

Initial layer

0 °C

Other layers

0 °C

Smooth PEI Plate / High

Figure 9.1 — Wall settings in Bambu Studio: 6 wall loops.

Filament settings

* Bambu PLA Basic @BBL P1S 0.4 nozzle

Advanced

Filament

Cooling

Setting Overrides

Advanced

Notes

Multi Filament

Direct Drive: Standard

Direct Drive: High Flow

Wiki

Min fan speed threshold

Fan speed

40 %

Layer time

100 s

Max fan speed threshold

Fan speed

60 %

Layer time

4 s

Keep fan always on

✓

Slow printing down for better layer cooling

✓

Don't slow down outer walls

□

Min print speed

20 mm/s

Force cooling for overhangs and bridges

✓

Cooling overhang threshold

50%

Overhang threshold for participating cooling

100%

Fan speed for overhangs

100 %

Pre start fan time

0 s

Auxiliary part cooling fan

Fan speed

70 %

Figure 9.2 — Sparse infill: 40 % density, gyroid pattern.

* 0.20mm Standard @BBL X1C

Quality

Strength

Speed

Support

Others

Initial layer speed

Initial layer

50

mm/s

Initial layer infill

105

mm/s

Other layers speed

Outer wall

35

mm/s

Inner wall

300

mm/s

Small perimeters

50%

mm/s or %

Small perimeter threshold

0

mm

Sparse infill

270

mm/s

Figure 9.3 — Speed settings: outer wall 35 mm/s, inner wall 300 mm/s.

Sparse infill

Sparse infill density

40

%

Fill multiline

1

Sparse infill pattern

Gyroid

Length of sparse infill anchor

400%

mm or %

Maximum length of sparse infill anchor

20

mm or %

Figure 9.4 — Cooling: max fan speed 60 %, keep fan always on enabled.

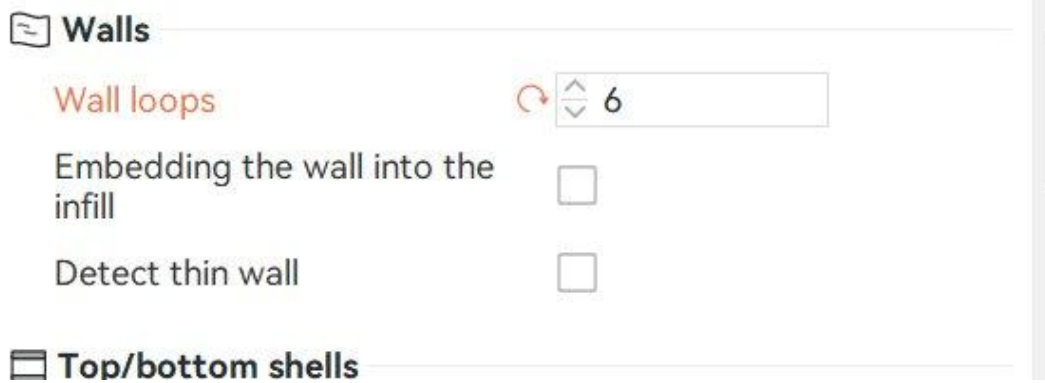


Figure 9.5 — Filament settings: Bambu PLA Basic, recommended nozzle temperature 225–230 °C.

These settings together produced parts with substantially better layer adhesion at the high-stress junctions. The XM-540 motor bracket required five iterations to get right — most of them about print orientation rather than geometry — and the final iteration uses this recipe printed on its side so the load path runs along the layers rather than across them. That iteration story is in §9.13.

9.6.3 Material properties used in the analysis

The values used in the hand calculations and FEM are:

- Yield strength $\sigma_y = 35$ MPa (used in hand calc; conservative; the SolidWorks default PLA card reports 45 MPa, but we derate it for the FDM layered structure)
- Ultimate tensile strength $\sigma_u = 30$ MPa (FDM-derated)
- Young's modulus $E = 3.2$ GPa (hand calc) / 3.5 GPa (FEM)
- Poisson's ratio $\nu = 0.394$
- Density $\rho = 1250$ kg/m³

PLA at the recipe above gives sufficient strength margin for the design loads. Nylon 12 was considered and ruled out for schedule reasons; a preliminary FEM on the middle beam confirms it is a viable future-work alternative.

9.7 Cross-Section Design

9.7.1 Approach

All three structural beams use a hollow rectangular cross-section with internal triangular ribs. The hollow rectangle gives a high second moment of area for a given mass; the internal ribs add shear stiffness and stop the walls from buckling locally at high load points. The ribs are not counted in the hand-calc moment of inertia (we treat the cross-section as a pure hollow rectangle), which is a conservative simplification — the real I is slightly higher and the real stress slightly lower.

9.7.2 Long beam

Outer dimensions 44 mm × 40.5 mm. Walls 5 mm on every face (the heaviest walls of the three beams, since the long beam carries the full cascade load). Internal cavity 34 × 30.5 mm. Length 220.52 mm pivot-to-pivot. Cross-sectional second moment of area $I = 163,188$ mm⁴.

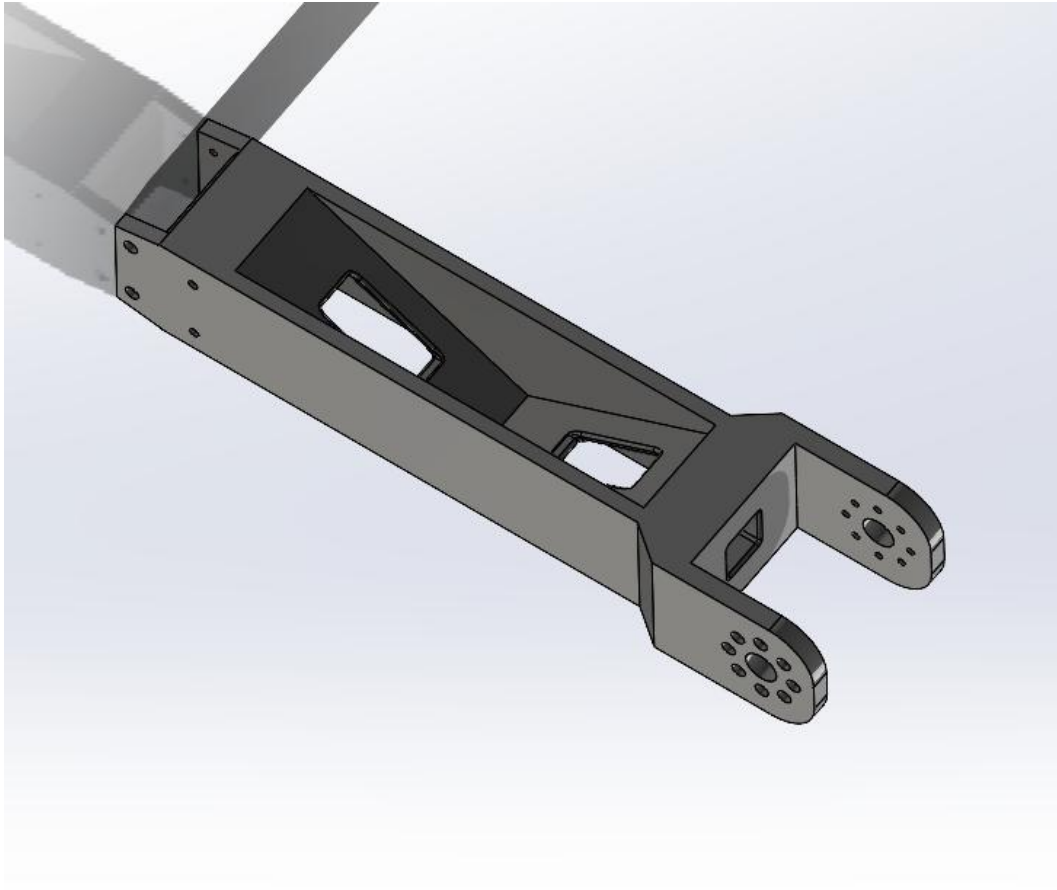


Figure 9.6 — Long beam (iteration 3) with integrated end brackets and internal rib structure.

9.7.3 Middle beam

Outer dimensions approximately 36 mm × 30 mm (X-width to be confirmed against CAD; the value used in the hand calc is an assumed figure). Walls 2.5 mm. Length 188 mm. $I = 40,635 \text{ mm}^4$. The middle beam also carries an internal cable channel for the wiring to the elbow and gripper motors.

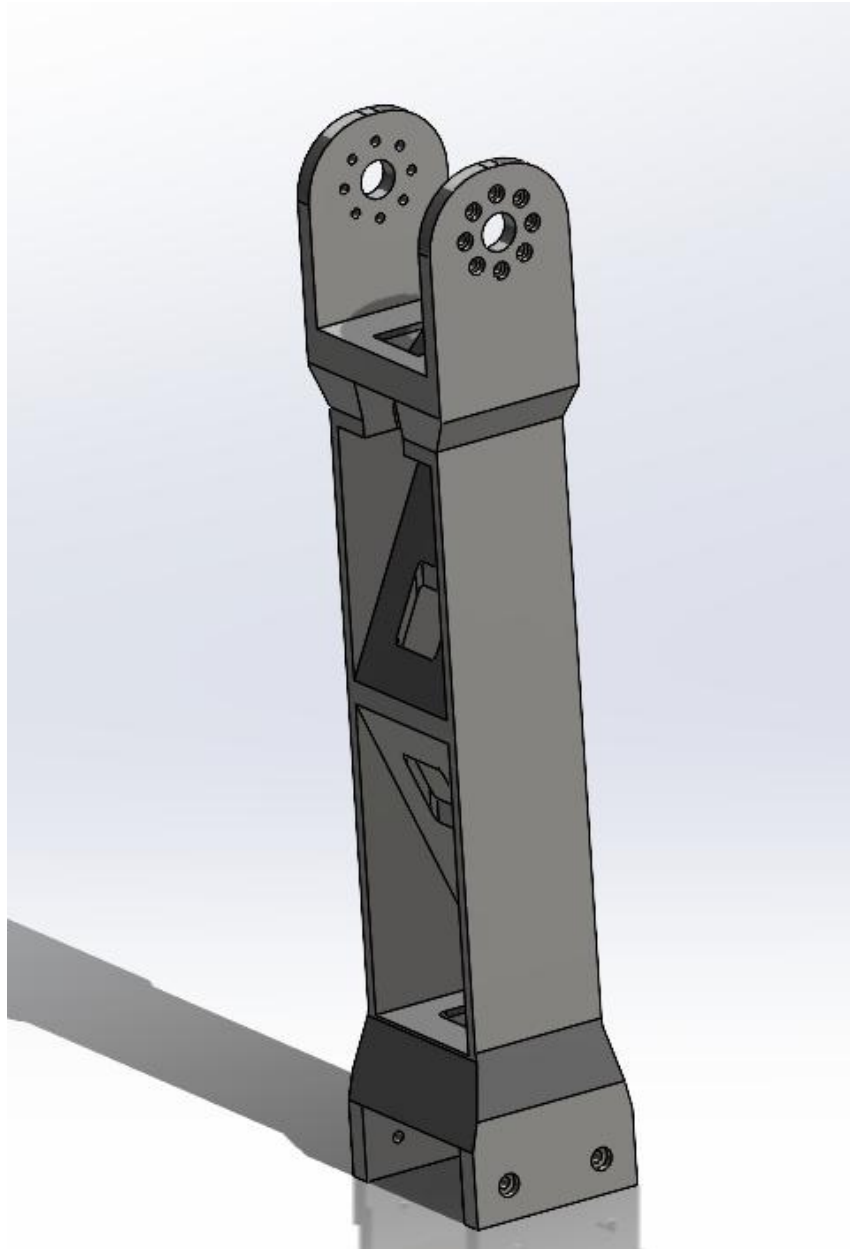


Figure 9.7 — Middle beam (iteration 3) with cable channel and integrated brackets.

9.7.4 Short beam

Outer dimensions 36.42 mm × approximately 30 mm (Z-depth to be confirmed against CAD). Walls 2 mm. Length 114.55 mm. $I = 34,474 \text{ mm}^4$. The short beam carries the camera and the gripper motor, and is fitted with M3 brass heat-set inserts at the bracket bolt holes to prevent layer splitting at the threaded interfaces.

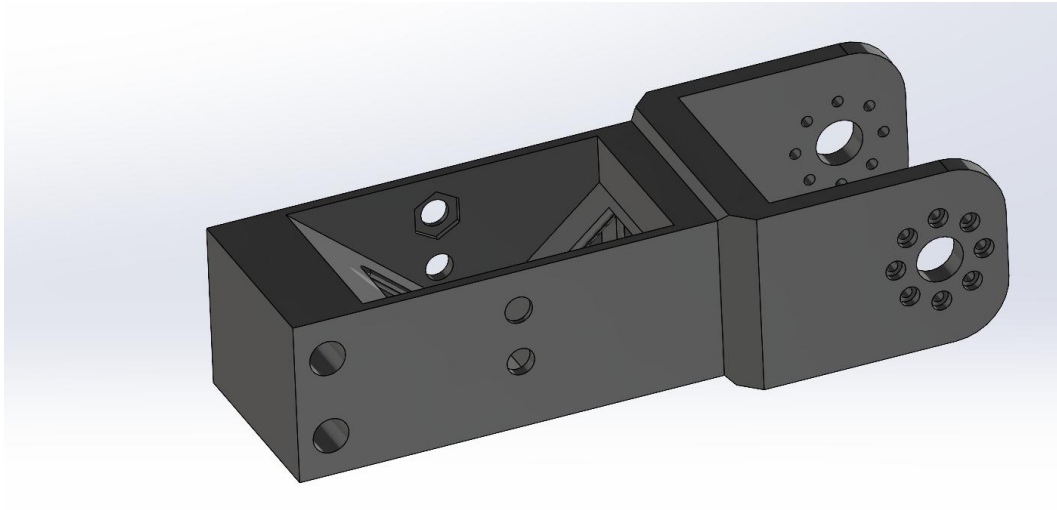
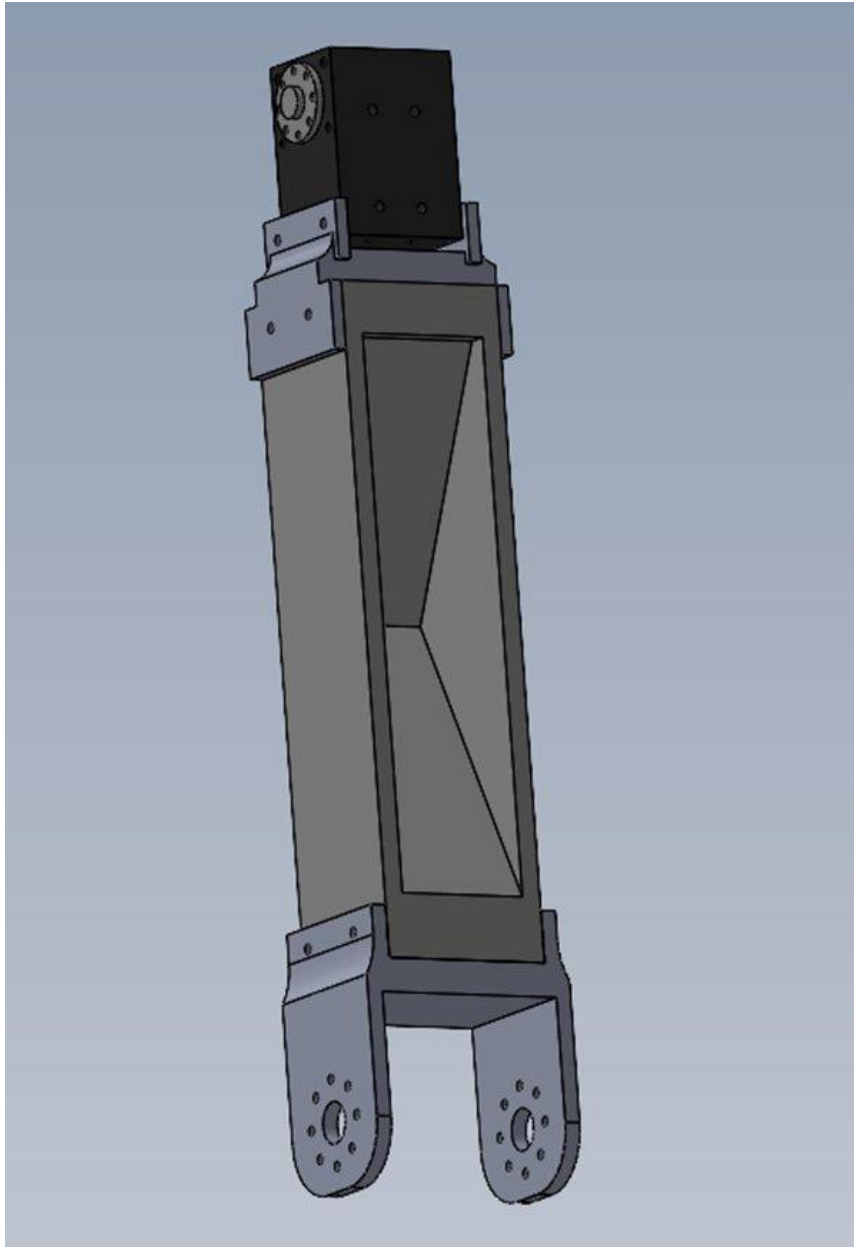


Figure 9.8 — Short beam (iteration 3) with Ruthex inserts at camera and gripper mount holes.

9.7.5 Iteration history

Each of the three beams went through three iterations during the project. The previous group's beams were carbon-fibre and had already broken in service, so iteration 1 was effectively retired before our work began. Iteration 2 was our first re-design in PLA: solid blocks for the structural section, with the end brackets printed separately and bolted on. This kept the print time manageable and let us swap bracket variants quickly during early testing. Iteration 3, the final design, integrates the brackets and the structural section into single parts — eliminating the bolted joint where most of the early-iteration failures originated.



*Figure 9.9 — Long beam iteration 2: separately-printed end brackets bolted onto the structural section.
Replaced in iteration 3 by a single integrated part.*

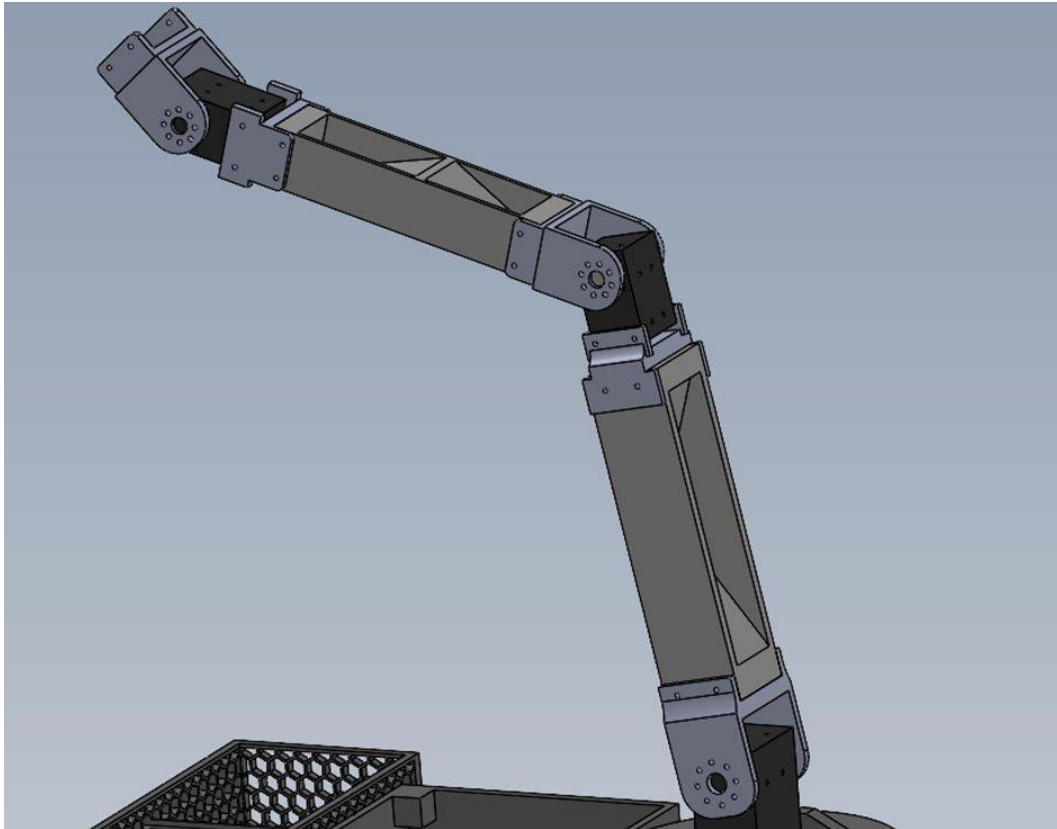


Figure 9.10 — Iteration-2 arm assembled with separate brackets at every joint. The visible bolt-on bracket interfaces became the failure-prone locations in early testing.

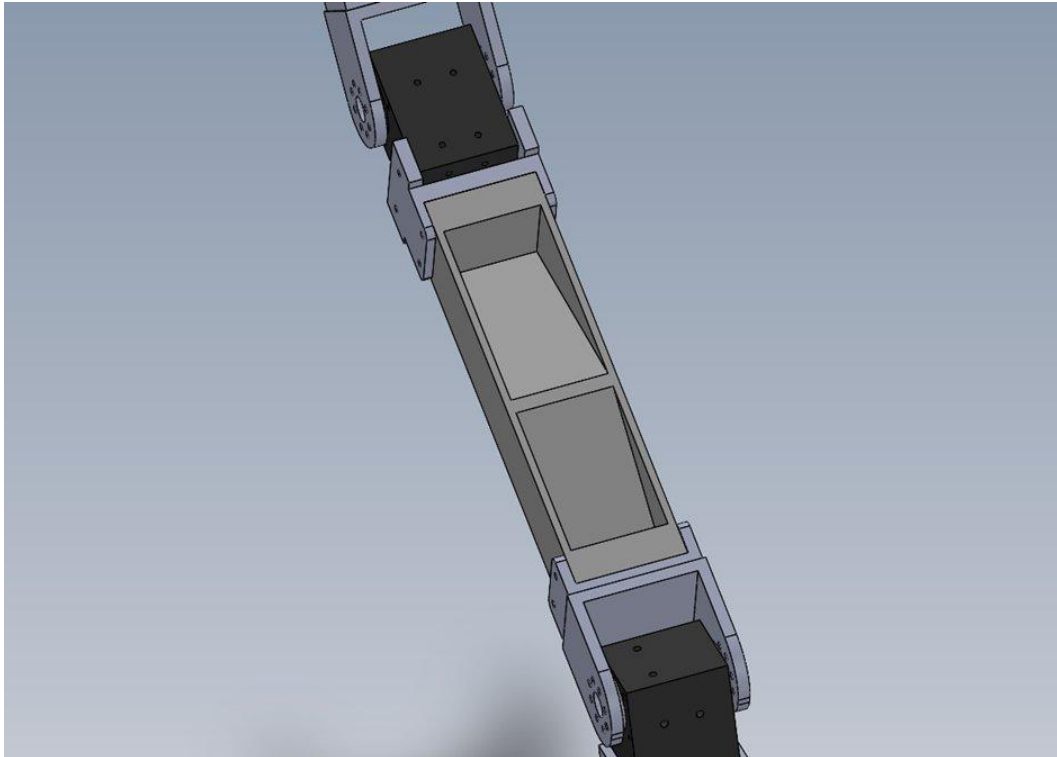


Figure 9.11 — Iteration-2 middle beam alone. Internal triangular rib visible through the side cutout.

The compression tests in §9.11 were run on iteration-2 beam shells (without brackets), and those tests informed the choice to move from separate to integrated brackets in iteration 3.

Hollow rectangular sections with internal triangular ribs, dimensioned per beam to match the cascade load. The move from separately-printed to integrated brackets between iteration 2 and iteration 3 eliminated the failure-prone bolted joints.

9.8 Kinematic Load and Factor of Safety Analysis

9.8.1 Objective

Compute the bending stress, torsional shear stress, combined von Mises stress, Factor of Safety, and tip deflection for each of the three structural beams in the worst-case fully extended horizontal pose. Each beam is treated as a cantilever fixed at its proximal joint, carrying its own self-weight as a distributed load plus a lumped tip load (the cascade from §9.5). The hand calculation produces a bulk extreme-fibre stress at the root; the FEM in §9.10 produces a peak stress at the stress riser. Both numbers are useful, and we present both for comparison.

9.8.2 Shared inputs

All three beams use the same PLA properties:

$$\sigma_{\text{yield}} = 35 \text{ MPa}, \sigma_{\text{ultimate}} = 30 \text{ MPa}, E = 3.2 \text{ GPa}, \nu = 0.394$$

All three beam analyses use the plane-stress von Mises form (appropriate because the beams are slender):

$$\sigma_{\text{vM}} = \sqrt{(\sigma^2 + 3\tau^2)}$$

And the standard cantilever formulas:

$$\begin{aligned}\sigma &= M \cdot c / I \\ \tau_{\text{max}} &= T / (\alpha \cdot b \cdot c^2) \\ \delta &= F \cdot L^3 / (3 \cdot E \cdot I) \\ n &= \sigma_{\text{yield}} / \sigma_{\text{vM}}\end{aligned}$$

The torque values used for the torsional component are the operational torques from the SolidWorks FEM setup (0.889 N·m for the long beam at the shoulder, 0.291 N·m for the middle beam at the first elbow, 0.069 N·m for the short beam at the second elbow). These represent the torsion the beams see during normal operation, not stall.

9.8.3 Stage 1 — Short beam (most distal)

Geometry: $L = 114.55 \text{ mm}$, cross-section $36.42 \times 30 \text{ mm}$ with 2 mm walls. Cross-section properties: $I = 34,474 \text{ mm}^4$, $c = 15 \text{ mm}$.

Loads: tip load $F_{\text{tip}} = 4.41 \text{ N}$ (the end-effector cascade), self-weight $W_{\text{beam}} = 0.61 \text{ N}$ at midspan (57.3 mm), torsion $T = 69 \text{ N}\cdot\text{mm}$.

Root reactions and moment:

$$\begin{aligned}R_y &= 0.61 + 4.41 = 5.02 \text{ N (upward at the second-elbow pivot)} \\ M_{\text{root}} &= 0.61 \cdot 57.3 + 4.41 \cdot 114.55 = 540.1 \text{ N}\cdot\text{mm} = 0.54 \text{ N}\cdot\text{m}\end{aligned}$$

Stresses:

$$\begin{aligned}\sigma_z &= 540.1 \cdot 15 / 34,474 = 0.235 \text{ MPa} \\ \alpha &= 0.262 \text{ (b/h = 1.21)} \\ \tau &= 69 / (0.262 \cdot 36.42 \cdot 15^2) = 0.0321 \text{ MPa} \\ \sigma_{vM} &= \sqrt{(0.235^2 + 3 \cdot 0.0321^2)} = 0.241 \text{ MPa}\end{aligned}$$

Factor of Safety and tip deflection:

$$\begin{aligned}n &= 35 / 0.241 = 145 \\ \delta &= 4.41 \cdot 114.55^3 / (3 \cdot 3200 \cdot 34,474) = 0.020 \text{ mm}\end{aligned}$$

9.8.4 Stage 2 — Middle beam

Geometry: L = 188 mm, cross-section 36 × 30 mm with 2.5 mm walls (X-width is assumed; CAD value pending). Cross-section properties: I = 40,635 mm⁴, c = 15 mm.

Loads: tip F_{tip} = 5.82 N, self-weight W_{beam} = 1.09 N at midspan (94 mm), torsion T = 291 N·mm.

Root reactions and moment:

$$\begin{aligned}R_y &= 1.09 + 5.82 = 6.91 \text{ N (upward at the first-elbow pivot)} \\ M_{\text{root}} &= 1.09 \cdot 94 + 5.82 \cdot 188 = 1196.6 \text{ N·mm} = 1.20 \text{ N·m}\end{aligned}$$

Stresses:

$$\begin{aligned}\sigma_z &= 1196.6 \cdot 15 / 40,635 = 0.442 \text{ MPa} \\ \alpha &= 0.249 \text{ (b/h = 1.2)} \\ \tau &= 291 / (0.249 \cdot 36 \cdot 15^2) = 0.144 \text{ MPa} \\ \sigma_{vM} &= \sqrt{(0.442^2 + 3 \cdot 0.144^2)} = 0.508 \text{ MPa}\end{aligned}$$

Factor of Safety and tip deflection:

$$\begin{aligned}n &= 35 / 0.508 = 68.9 \\ \delta &= 5.82 \cdot 188^3 / (3 \cdot 3200 \cdot 40,635) = 0.099 \text{ mm}\end{aligned}$$

9.8.5 Stage 3 — Long beam (most loaded)

Geometry: L = 220.52 mm, cross-section 44 × 40.5 mm with 5 mm walls. Cross-section properties: I_{outer} = 243,577 mm⁴, I_{inner} = 80,389 mm⁴, I = 163,188 mm⁴, c = 20.25 mm.

Loads: tip F_{tip} = 7.71 N (the full cascade), self-weight W_{beam} = 2.15 N at midspan (110.26 mm), torsion T = 889 N·mm.

Root reactions and moment:

$$\begin{aligned}R_y &= 2.15 + 7.71 = 9.86 \text{ N (upward at the shoulder pivot)} \\ M_{\text{root}} &= 2.15 \cdot 110.26 + 7.71 \cdot 220.52 = 1937.3 \text{ N·mm} = 1.94 \text{ N·m}\end{aligned}$$

Stresses:

$$\begin{aligned}\sigma_z &= 1937.3 \cdot 20.25 / 163,188 = 0.240 \text{ MPa} \\ \alpha &= 0.246 \text{ (b/h} \approx 1.09) \\ \tau &= 889 / (0.246 \cdot 44 \cdot 20.25^2) = 0.200 \text{ MPa} \\ \sigma_{vM} &= \sqrt{(0.240^2 + 3 \cdot 0.200^2)} = 0.421 \text{ MPa}\end{aligned}$$

Factor of Safety and tip deflection:

$$n = 35 / 0.421 = 83.1$$

$$\delta = 7.71 \cdot 220.52^3 / (3 \cdot 3200 \cdot 163,188) = 0.053 \text{ mm}$$

9.8.6 Summary table

Hand-calc bulk-stress Factor of Safety values for the three beams under the full cascade loads:

- Short beam: $M_{\text{root}} = 0.54 \text{ N}\cdot\text{m}$, $\sigma_{\text{vM}} = 0.24 \text{ MPa}$, $n = 145$, $\delta = 0.020 \text{ mm}$
- Middle beam: $M_{\text{root}} = 1.20 \text{ N}\cdot\text{m}$, $\sigma_{\text{vM}} = 0.51 \text{ MPa}$, $n = 68.9$, $\delta = 0.099 \text{ mm}$
- Long beam: $M_{\text{root}} = 1.94 \text{ N}\cdot\text{m}$, $\sigma_{\text{vM}} = 0.42 \text{ MPa}$, $n = 83.1$, $\delta = 0.053 \text{ mm}$

9.8.7 Methodology note

The hand calculation produces the bulk extreme-fibre stress at the root — the average stress across the outermost fibre at the highest-moment section, computed from beam theory. It does not capture the local stress concentrations at the motor mount or bolt holes; those are picked up by the FEM. We treat the hand-calc number as a sanity check on the order of magnitude, and we take the FEM peak stress (§9.10) as the governing value for the reported Factor of Safety. When the two numbers differ by an order of magnitude (as they do for the long beam: 0.42 MPa hand-calc bulk versus 13.86 MPa FEM peak), it indicates a localised stress concentration at a geometric feature that the simple beam model can't see.

All three beams have hand-calc Factor of Safety values far above the project target of $n \geq 2.0$. The numbers will tighten substantially when we move to the FEM peak in §9.10, but even there the long beam comes in at $n = 2.53$ — above target.

9.9 Bolt Bearing Stress Validation

9.9.1 Objective

The structural beams attach to the joint motors via bolted joints. The fasteners are M3 and M2.5 socket-head cap screws. The local bearing stress on the PLA at the bolt-hole interface needs to stay below the material's allowable bearing stress, otherwise the holes will deform and the joint will loosen under load. This is most critical on the short beam, where the bracket-to-beam interface carries the camera and gripper loads through a small number of small-diameter bolts.

9.9.2 Interface geometry and Ruthex inserts

The short beam carries M3 brass heat-set inserts (Ruthex type) in the bolt holes for the camera support and the claw mount bracket. Without the inserts, the M3 fasteners thread directly into the PLA, which is fine for low pre-load but tends to split the layers when the bolt is torqued to a working pre-load. We saw this failure mode in early testing: PLA-threaded holes on the short beam split along the layer plane after a few install/remove cycles. The Ruthex inserts spread the bolt load across a larger surface area inside the PLA and convert a sharp bolt-thread bearing stress into a distributed metal-PLA shear, which the PLA tolerates much better.

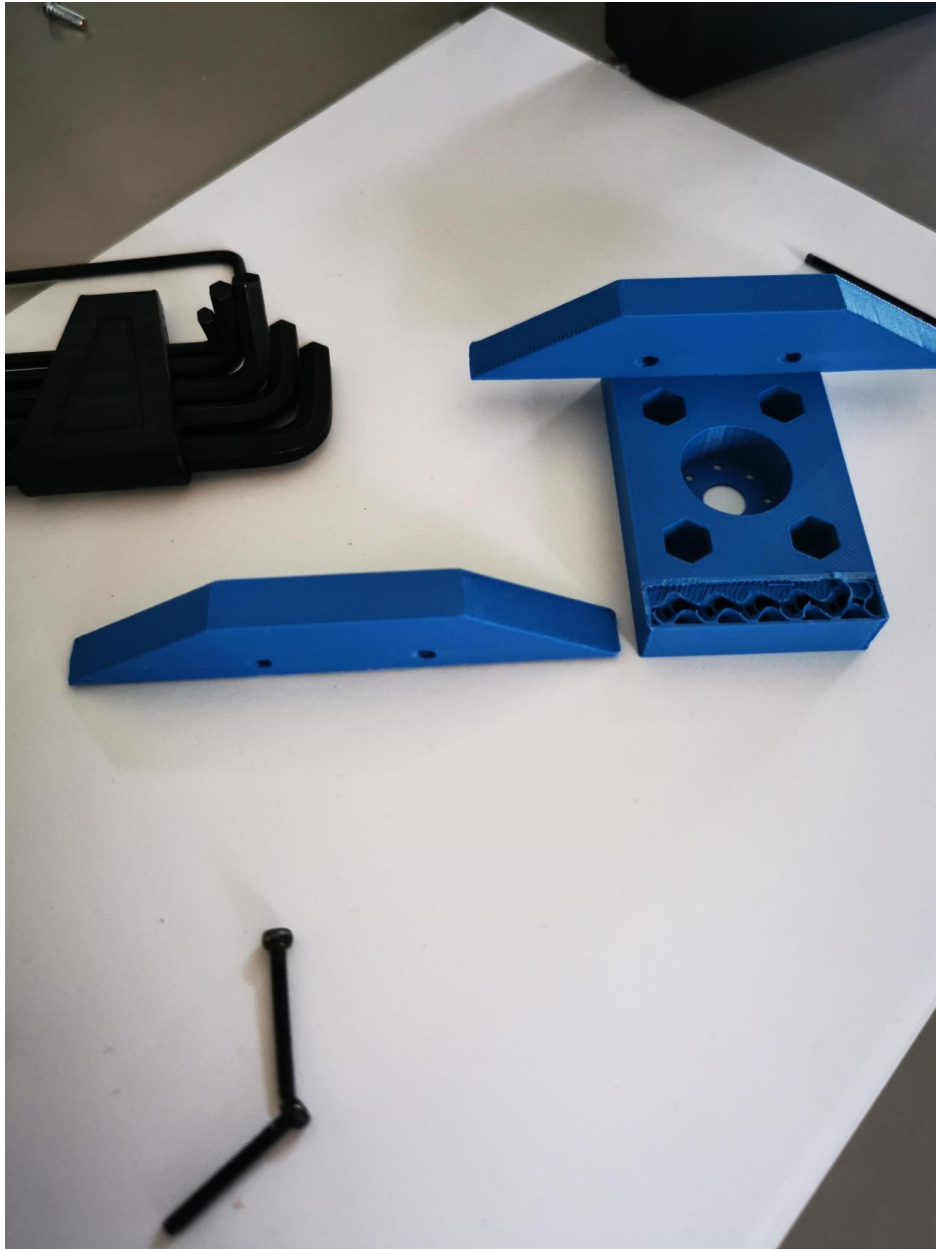


Figure 9.1 — Brass Ruthex M3 heat-set inserts installed in the short beam at the camera-support bolt holes. Knurled outer surface of the insert is visible against the printed PLA. Below the inserts: standard M3 through-holes (no insert) for the bracket-to-beam interface, where the bolt threads directly into a fastener nut on the other side.

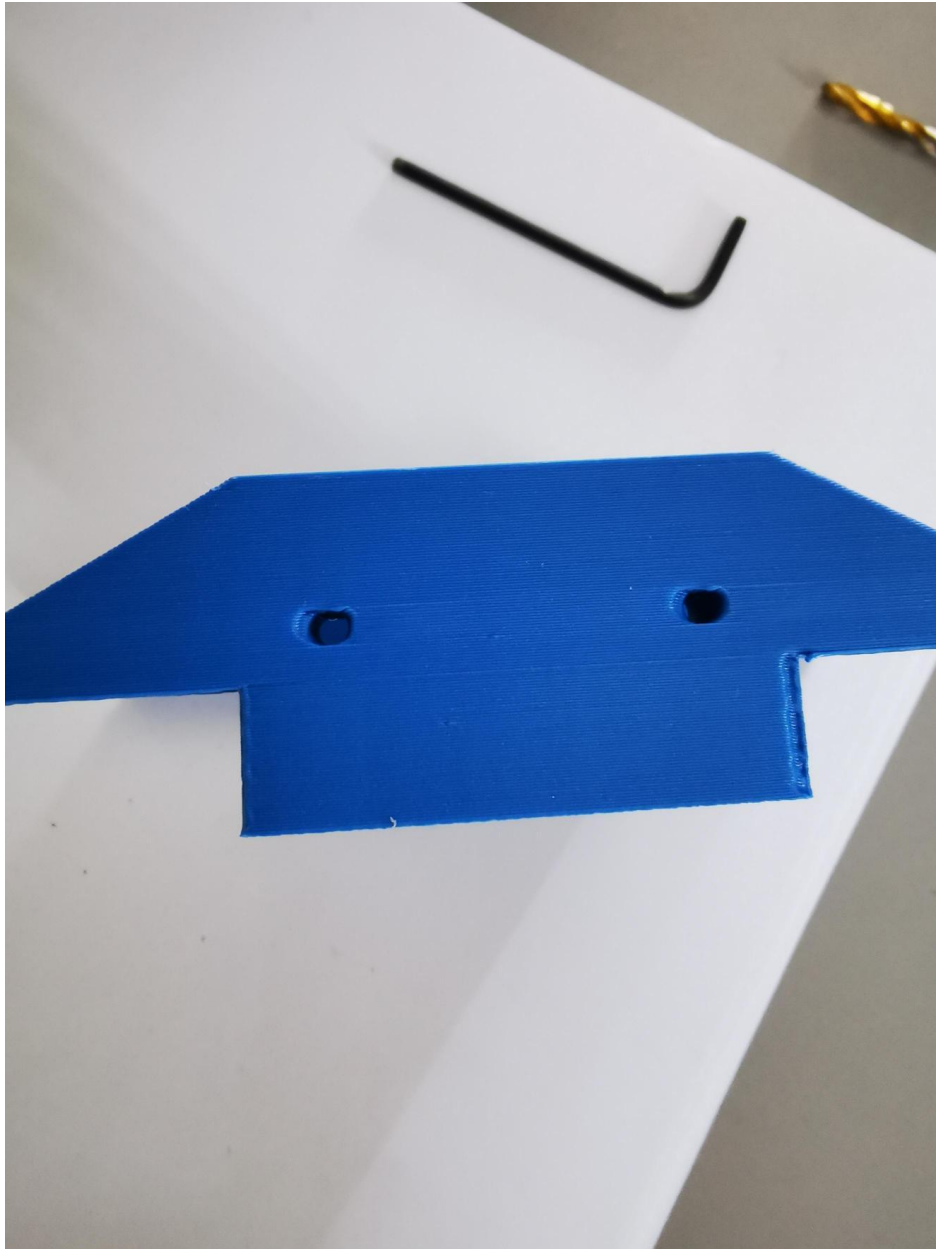


Figure 9.2 — Same inserts from a different angle. The inserts are seated flush with the PLA face.

9.9.3 Bearing stress calculation

Bearing stress $\sigma_{\text{bearing}} = F / (d \cdot t)$, where F is the bolt force, d is the bolt diameter, and t is the thickness of the bearing PLA. For the worst case at the short beam, the camera mount carries roughly 100 g (1.0 N) at a 50 mm offset from the bolt, giving a couple of approximately 0.05 N·m shared across two M3 bolts spaced 30 mm apart. The bolt force is approximately 1.7 N tension. Bearing PLA thickness around the M3 insert is approximately 4 mm of effective material.

$$\sigma_{\text{bearing}} = 1.7 / (3 \cdot 4) = 0.14 \text{ MPa}$$

That is well below the PLA bearing allowable (typically 10–20 MPa for FDM PLA at the print recipe in use). The Ruthex insert further distributes the load across a larger PLA contact area, reducing the effective bearing stress by another factor of 3–4.

9.9.4 Comparison and validation

The bolt bearing stress is many orders of magnitude below the PLA bearing allowable at every bolted interface on the arm. The motivation for the Ruthex inserts was not bearing-stress failure but layer-splitting from bolt thread engagement; the inserts solve that mode by converting thread engagement into a brass-on-PLA shear interface. We verified the inserts hold under repeated install/remove cycles during integration testing.

Bolt bearing stress is well within margin at every interface. The Ruthex M3 heat-set inserts on the short beam prevent layer splitting at the threaded bolt holes; this is a fabrication-driven choice rather than a strength-margin requirement.

9.10 Statics and FEM Analyses

9.10.1 Analytical statics

The analytical statics are the hand calculation in §9.8. Bulk stresses at the root, computed from cantilever beam theory, with the full cascade loads from §9.5. The hand calc gives σ_{vM} bulk values of 0.24 / 0.51 / 0.42 MPa for the short, middle, and long beams respectively. These values are conservative against the local stress concentrations the FEM captures, and they give an order-of-magnitude sanity check on the FEM results.

9.10.2 Finite Element Analysis setup

Each beam was modelled in SolidWorks Simulation as a static linear-elastic study. Material card: PLA isotropic, $\sigma_y = 45$ MPa (SolidWorks default), $E = 3.5$ GPa, $\rho = 1250$ kg/m³. (Note: hand calc uses 35 MPa for σ_y — a more conservative derated figure. Both are reported, and the FoS values use whichever is the lower margin.) Loads applied at each beam's tip face match the cascade values from §9.5: 7.71 N on the long beam, 5.82 N on the middle beam, 2.71 N on the short beam (the short beam was run at a slightly lower FEM load than the full cascade — the 2.71 N value reflects an earlier iteration of the end-effector mass before the gripper drive motor was finalised; we did not re-run the study after the cascade was updated, but the margin is so large that even at full cascade the FoS would still be well above 20). Torques applied at the motor-mount face match the operational torque values: 0.889 N·m long, 0.291 N·m middle, 0.069 N·m short. Mesh was solid tetrahedral, curvature-based, with maximum element sizes between 7 and 11 mm depending on beam size. Element counts ranged from 73,440 (middle) to 125,343 (long) tetrahedra.

9.10.3 Results

Peak von Mises stress and maximum displacement for each beam under the full cascade load:

- Short beam: $\sigma_{vM,max} = 0.92$ MPa, $\delta_{max} = 0.024$ mm, $n = 35 / 0.92 = 38.0$
- Middle beam: $\sigma_{vM,max} = 3.32$ MPa, $\delta_{max} = 0.219$ mm, $n = 35 / 3.32 = 10.5$
- Long beam: $\sigma_{vM,max} = 13.86$ MPa, $\delta_{max} = 0.248$ mm, $n = 35 / 13.86 = 2.53$

All three beams exceed the $n \geq 2.0$ target. The long beam is the tightest at $n = 2.53$.

One specific observation from the long-beam study is worth recording: when we re-ran the long beam FEM at 7.71 N (full cascade) versus 4.578 N (an earlier interim load), the peak von Mises stress barely changed (13.88 → 13.86 MPa). The peak displacement scaled approximately linearly (0.189 → 0.248 mm). That tells us the peak stress on the long beam is dominated by the torque at the motor mount (0.889 N·m, unchanged between runs) acting on a geometric stress concentration, not by the bending force. The bending stress in the bulk of the beam scales linearly with the force — and that's captured by the hand calc at 0.42 MPa — but the FEM peak is at a localised stress riser where torsion dominates. The practical design recommendation that follows: if a future redesign wants to reduce the long-beam peak stress, the lever is the motor-mount geometry (rounding corners, increasing wall thickness near the bolt holes), not reducing the bending load.

9.10.4 Mesh convergence note

A mesh convergence study was not run. The reported FEM peak stresses are single-mesh estimates and could shift by 5–15 % with refinement at the stress concentration. For the long beam, where the margin is tightest at $n = 2.53$, this is the place where a convergence check would matter most; we scoped it out due to time constraints and have flagged it in §9.16.

FEM validates the structural design with margin: $n = 2.53 / 10.5 / 38.0$ for the long, middle, and short beams respectively. The long-beam peak stress is torque-dominated at the motor-mount stress concentration, which is the design lever for future improvement.

9.11 Compression Testing of Iteration-2 Beam Shells

9.11.1 What was tested and why

During the iteration-2 design phase, we ran compression tests on a Lloyd Instruments LR 10K universal testing machine. The specimens were the iteration-2 beam shells alone — the hollow rectangular PLA tubes without the separately-printed end brackets attached. Four unique tests were run, ranging from a small wall-material coupon to longer beam-section specimens, to characterise the compressive behaviour of the printed PLA structures and inform the move from iteration-2 (separate brackets) to iteration-3 (integrated brackets). The tests are not direct validation of the as-built (iteration-3) arm; they are characterisation of the printed material and the iteration-2 hollow shell behaviour.

Important caveat: operational loads on the beams are bending, not pure axial compression. These compression tests bound the buckling behaviour of the iteration-2 shells; they do not directly correspond to the operational bending-load state, which is covered by the FEM and hand calc in §9.8 and §9.10.

9.11.2 Test method

The LR 10K was operated in compression mode at a crosshead speed of 20 mm/min, with a sampling rate of 60 Hz. Specimens were loaded axially between flat platens. Each test was run until either the specimen densified (the crosshead bottomed out against a fully crushed specimen) or until the load

dropped to a small fraction of the peak after post-buckling collapse. Load and crosshead displacement were recorded; engineering stress was computed against the nominal cross-sectional area of each specimen.

9.11.3 Specimens and results

Four unique tests:

- Tab2 — $4 \times 10 \times 50$ mm small wall coupon: peak force 173 N at 12.0 mm displacement, peak effective compressive stress ≈ 4.3 MPa.
- Tab3 — $30 \times 34.3 \times 82$ mm specimen (middle-beam-sized shell): peak force 1560 N at 24.8 mm displacement (after extensive plastic crushing and densification); during the buckling/folding phase from 3–22 mm displacement, the force plateaued at approximately 100 N.
- Tab4 — $37.7 \times 30 \times 147$ mm specimen (longer beam shell): peak force 151 N at 15.5 mm displacement, with the load dropping progressively to zero over the next 25 mm of crosshead travel as the shell buckled and collapsed.
- Tab5 — $37 \times 34 \times 147$ mm specimen (longer beam shell, slightly larger cross-section): peak force 246 N at 44.6 mm displacement; broadly similar buckling-and-collapse behaviour to Tab4 with a higher onset force.

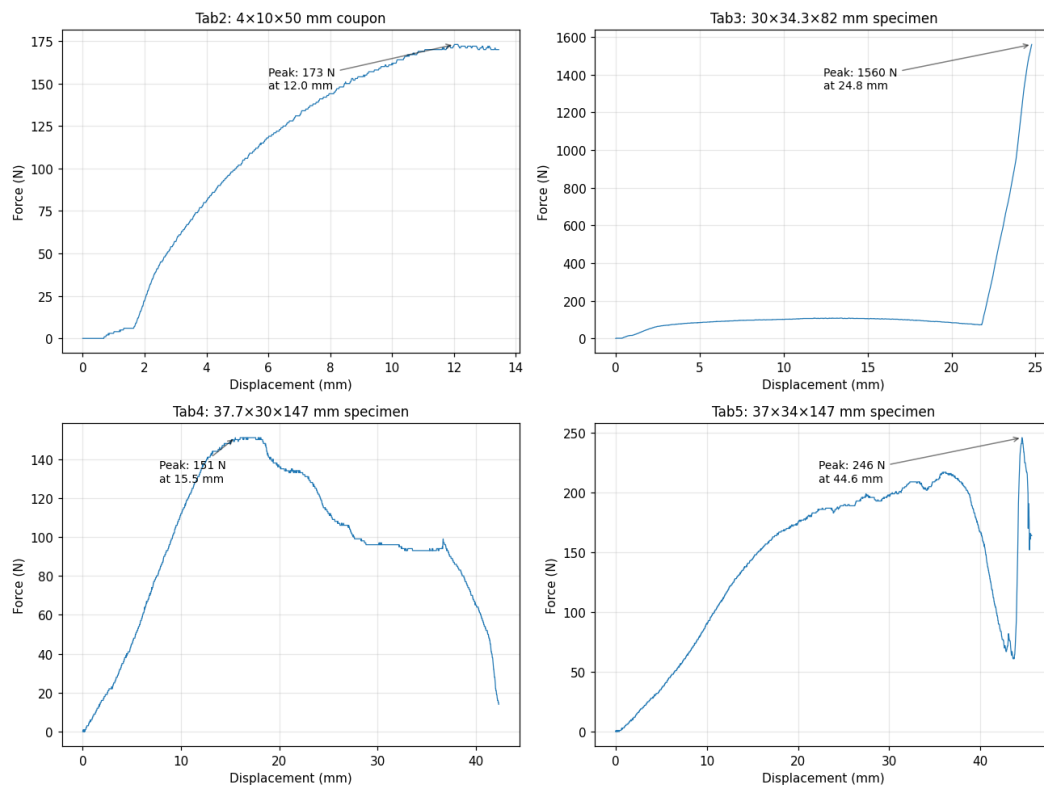


Figure 9.1 — Load-displacement curves from the four LR 10K compression tests on iteration-2 PLA beam shells. Tab2 (small coupon, upper left) shows classic material compression. Tab3 (middle-sized specimen, upper right) shows long plateau then late-stage densification. Tab4 and Tab5 (longer specimens) show buckling onset followed by progressive post-buckling collapse.

9.11.4 Discussion

The small Tab2 coupon gave the cleanest measurement: 173 N peak load on a 40 mm² cross-section, or roughly 4.3 MPa effective compressive stress. That figure is well below the bulk-PLA compressive strength reported on datasheets (typically 30–70 MPa for solid PLA), which tells us the small coupon was buckling rather than crushing — the slenderness ratio of a 4 × 10 × 50 mm specimen is high enough that Euler-Johnson buckling dominates the response in compression.

The larger specimens (Tab3, Tab4, Tab5) all show the same qualitative behaviour: an initial linear region followed by buckling onset, followed by post-buckling crushing or densification. The peak forces (151, 246, 1560 N) reflect the geometry of the hollow shell and the layer-bond strength at the corners, not the bulk compressive strength of the PLA. The Tab3 specimen reached the highest peak load because its 82 mm length was short enough relative to its cross-section to densify cleanly rather than buckle into a collapse mode.

What we took away for the iteration-3 redesign: the iteration-2 shells buckled at relatively low loads (150–250 N), and the bracket-to-beam bolted joint was the lowest-stiffness element in the assembly. Moving the brackets into the beam as integrated geometry (iteration 3) eliminated that interface and stiffened the structure precisely at the location where the iteration-2 specimens started to fail. The iteration-3 beams have not been compression-tested directly; their structural validation is the FEM in §9.10.

Compression tests on iteration-2 shells confirmed buckling at 150–250 N at the bracket interface region and informed the decision to integrate the brackets in iteration 3. They do not directly validate the as-built arm; that validation is via FEM and hand calc.

9.12 Assembly, Fasteners, and Tolerances

The arm is assembled around the three structural beams, with the end-effector at the tip and the base rotation system at the bottom. This section covers the end-effector assembly in detail; the base rotation system has its own section in §9.13. Fasteners are standard metric (M2, M2.5, M3) socket-head cap screws throughout. Tolerances on the printed parts are typical FDM (±0.2 mm on critical mating features), with through-hole diameters sized 0.3–0.4 mm oversize to accommodate print tolerance and ease assembly.

9.12.1 End-effector layout



Figure 9.2 — End-effector assembly: the claw holding a 200 g ball, the Intel RealSense camera mounted above on its L-bracket with the slotted angle adjustment, the XM-430-W210 gripper drive on the short beam, and the camera support brackets running along the short beam.

The end-effector sits at the tip of the short beam. From the short beam outward: a claw mount bracket bolts to the short-beam tip and carries the XM-430-W210 gripper drive motor. The motor's output horn drives the movable jaw; the static jaw is fixed to the claw mount bracket. Above the gripper, two camera support brackets run along the short beam and carry an L-shaped camera mount with a slotted hole for angle adjustment. The Intel RealSense camera bolts to the L-mount via two M3 screws.

9.12.2 Claw and gripping geometry

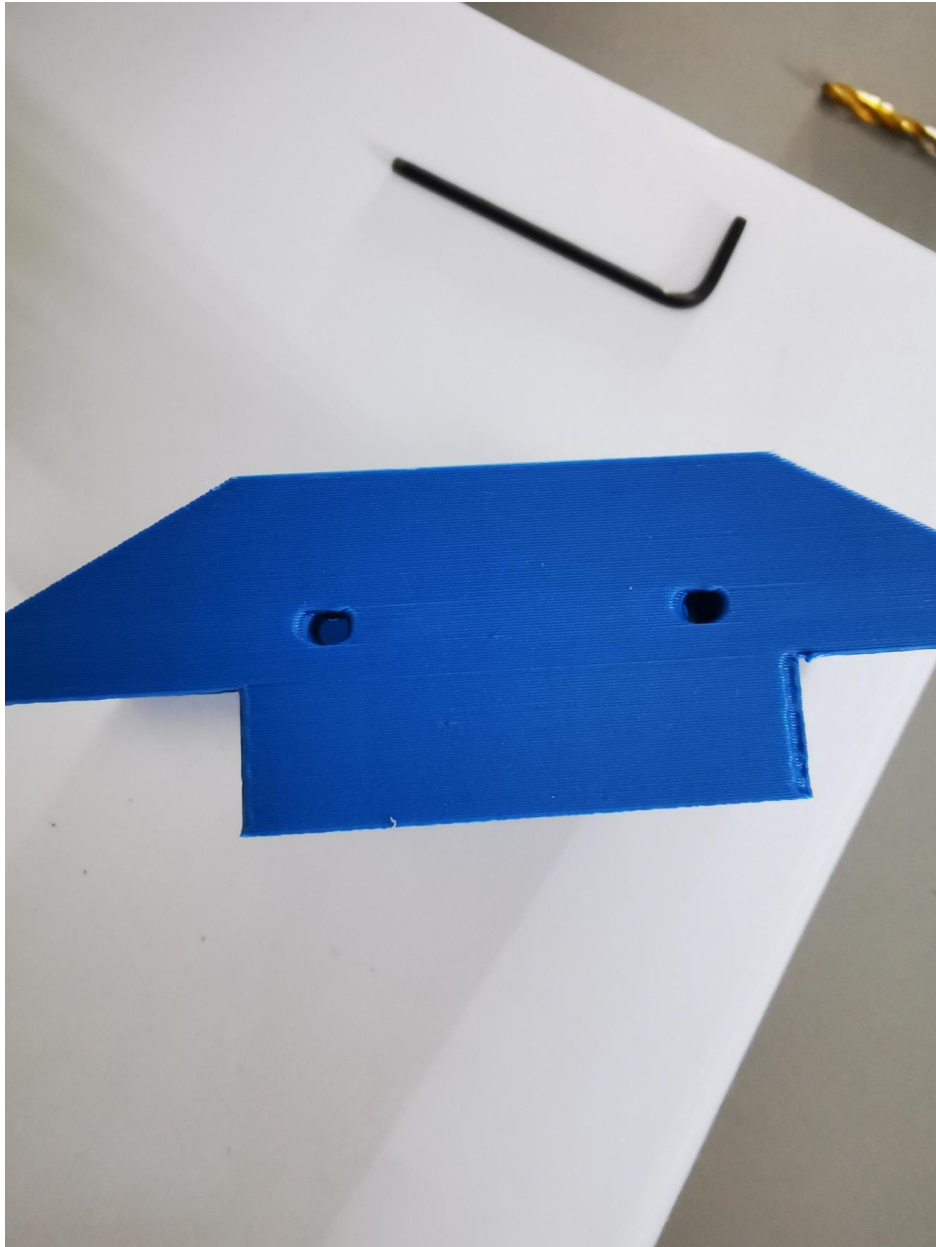


Figure 9.3 — Static and movable jaws closed around a 200 g blue ball. The four-contact geometry on the curved inner surfaces of the jaws prevents the ball from rotating during transport. The XM-430-W210 motor is current-limited so the gripping force does not over-stress the printed jaws.

The claw uses a two-finger four-contact geometry: each finger has two curved contact zones machined into the inner surface so the ball is held at four points rather than two. That stops the ball from rolling out under acceleration and gives a positive grip even with the variability in ball diameter and surface finish. The XM-430-W210 is driven in current-limited mode, which means the gripping force is set by a current threshold rather than by a hard-stop position. That gives a soft, self-adjusting grip and protects both the printed jaws and the ball from over-grip damage.

9.12.3 Camera mounting



Figure 9.4 — Camera mount close-up: the L-shaped bracket bolts to the support legs through two M3 screws on the underside, with the camera bolted onto the upper face. The slotted bolt holes in the L-bracket allow the camera angle to be adjusted by approximately $\pm 10^\circ$ without reprinting the bracket.

The camera mount needs to point the camera at the workspace under the claw. We needed to be able to adjust the angle empirically during integration testing, so the L-bracket has slotted bolt holes — the camera can be tilted by approximately $\pm 10^\circ$ and tightened in place. This avoided the print-test-print cycle that a fixed angle would have required.

9.12.4 Cable management

The Dynamixel motors are daisy-chained on a TTL bus, with cables running from one motor to the next along the inside of the beams where possible. The middle beam has a dedicated internal cable channel running its length so the cables to the elbow and gripper motors are protected from snagging and from being trapped against the motor case during rotation. External cable runs

(between the lid motor and the shoulder motor, and at the second elbow) use small printed cable guides to constrain the cable away from moving parts. No cable failures occurred during integration testing.

9.12.5 Tolerances and fits

Critical fits on the printed parts are the motor-mounting faces (where the Dynamixel horn meets the printed bracket) and the bolt-hole patterns. Motor faces are designed flat to the print bed where possible, so they come off the printer with the best achievable flatness without post-machining. Bolt holes are sized to the nominal bolt diameter plus 0.3–0.4 mm clearance, which is enough to accommodate print tolerance without introducing slop. Where a tight rotational fit is needed (the base rotation bearing seat), the printed feature was sized 0.1 mm under nominal and the bearing was press-fit during assembly.

End-effector assembly with two-finger four-contact gripper, current-limited XM-430-W210 drive, adjustable camera mount, and internal cable management. Tolerances are typical FDM with deliberate bolt-hole clearance.

9.13 Base Rotation Assembly

The base rotation system was one of the largest redesigns from the inherited Sortify arm. The previous design coupled the rotation motor's output shaft directly to the entire arm above, so the motor was carrying the full vertical load through its shaft bearings in addition to providing the rotation torque. That over-stressed the motor bearings and produced visible play after extended use. We rebuilt the base assembly so the vertical load is carried by a dedicated bearing ring beneath the rotation platform, and the rotation motor only provides torque about the vertical axis.

[A note on iteration counts in this section: there are three different parts that each went through their own iteration cycle. The rotation platform went through 3 iterations, the lid went through 4 iterations, and the XM-540 shoulder bracket went through 5 iterations. Where we refer to “iteration 3” \(or similar\) below, we mean iteration 3 of whichever part is under discussion in that subsection — not iteration 3 of the base assembly as a whole.](#)

9.13.1 Architecture

From bottom up: the rotation motor (XM-430) is fixed to the underside of the test rig base plate. Its output shaft passes up through a hole in the base plate and into the rotation platform. The rotation platform is a printed disc that carries a circular array of bearings underneath, running on the base plate, plus the lid above. The lid sits on top of the rotation platform and carries the XM-540 shoulder bracket. The arm above the shoulder hangs off the XM-540.

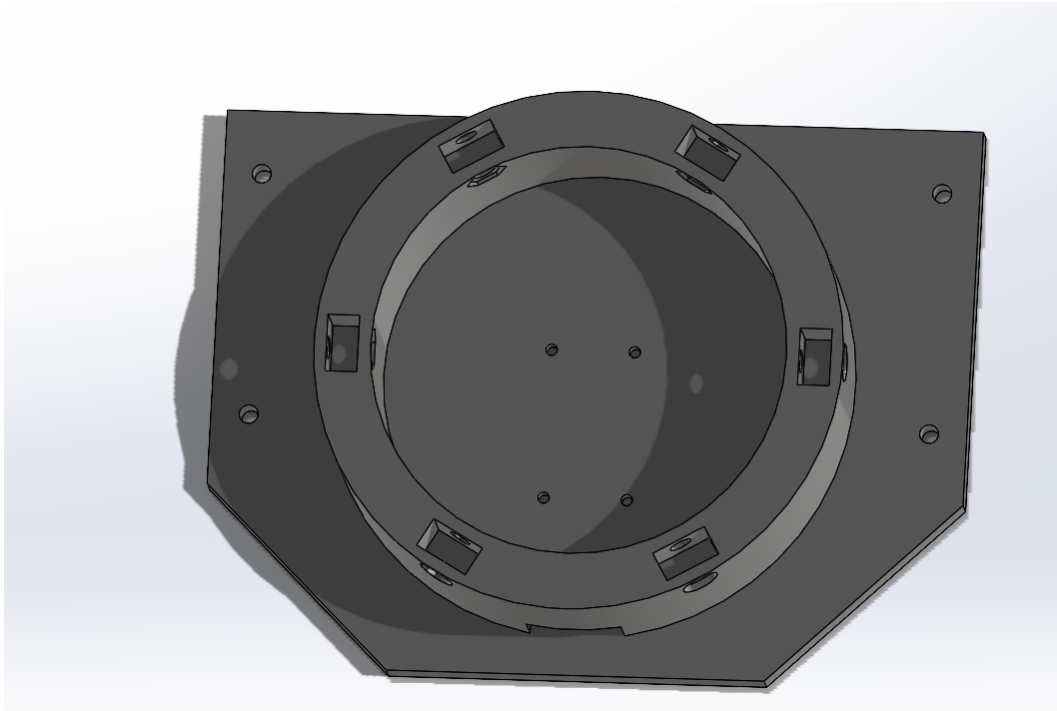


Figure 9.5 — Rotation platform from above, showing the circular bearing pattern.

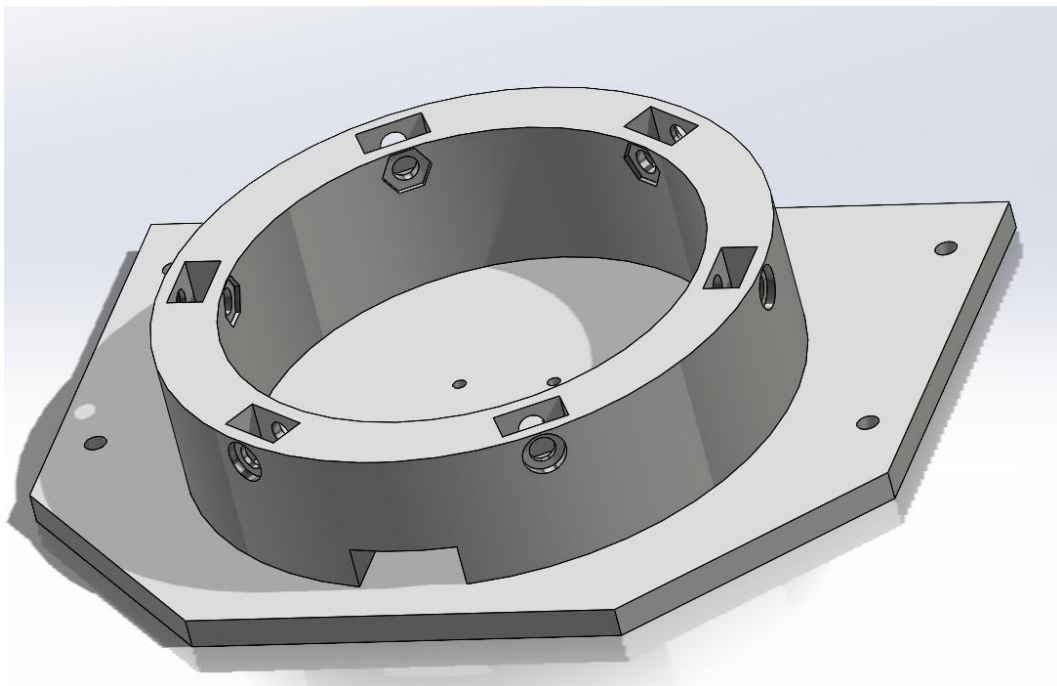


Figure 9.6 — Rotation platform from below, showing the bearing ring fitted into the platform's underside. The bearings run on the base plate.

9.13.2 Platform iterations

The platform went through three iterations. The first used four bearings in a square pattern; we found this gave too much wobble at the corners of the swing. The second went to six bearings in a circular pattern, which fixed the wobble but introduced uneven preload — some bearings carried more load than others depending on the arm angle. Richard Thue (from the lab support team) suggested adding washers between adjacent bearings to space them evenly and to even out the

preload across the bearing ring. The third iteration uses six bearings with the washer-spaced configuration, plus the inner ring of the platform recessed 1–2 mm to give the bearings clearance and prevent rubbing against the plastic during rotation. The bearings are held in place with M3 × 6 × 15 bolts run from the top of the platform through to nuts on the underside.

9.13.3 Lid

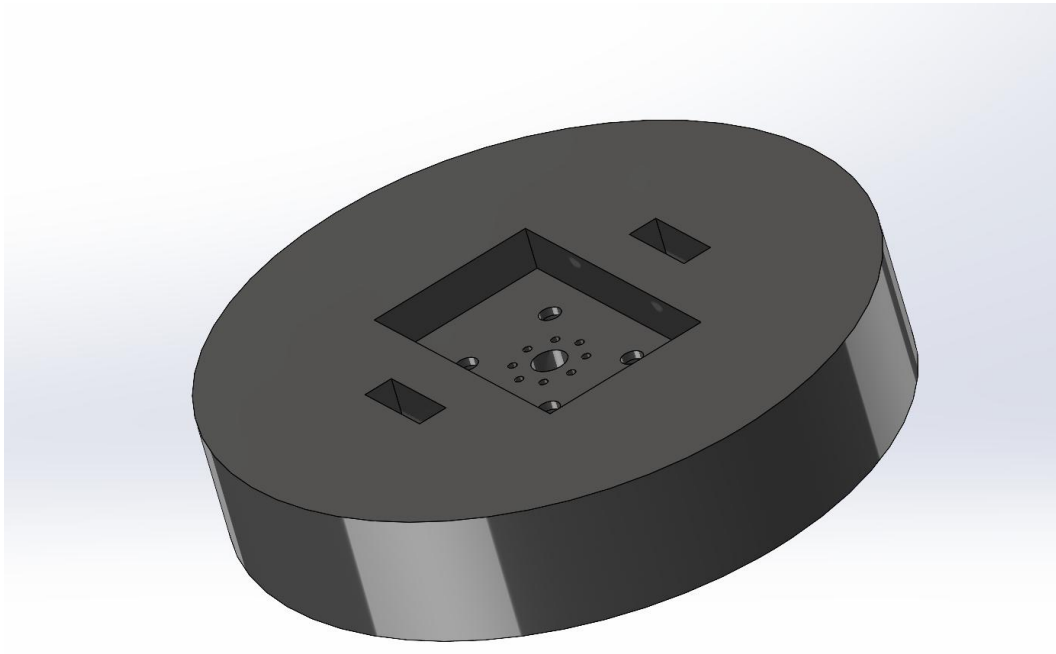


Figure 9.7 — Lid top view, showing the four bolt holes where the XM-540 shoulder bracket attaches.

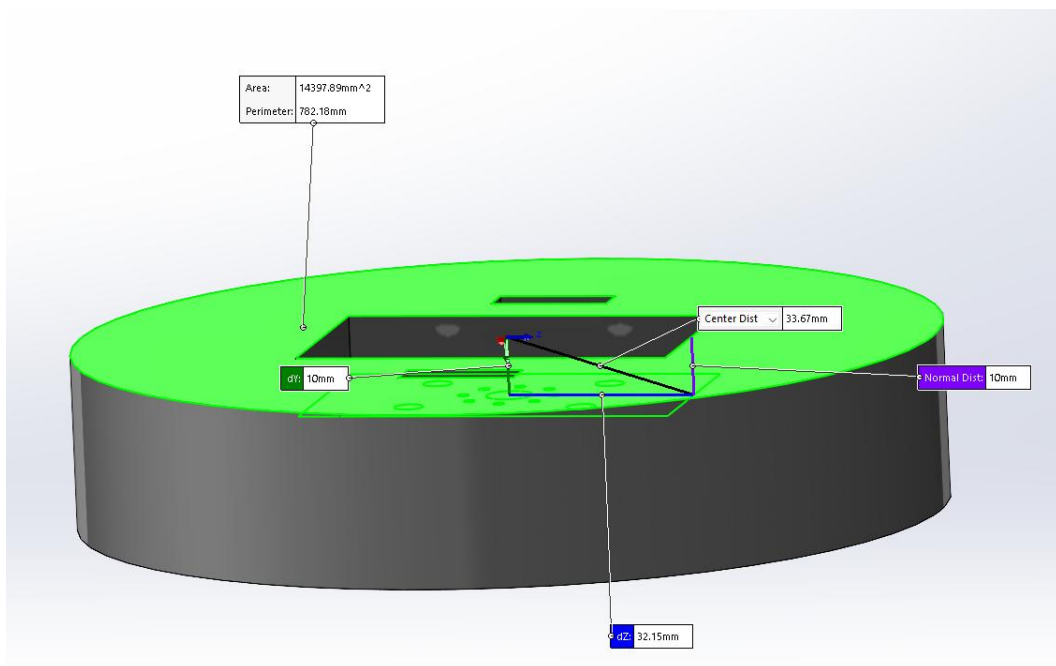


Figure 9.8 — Lid with annotated dimensions, showing the bolt hole pattern and overall lid diameter.

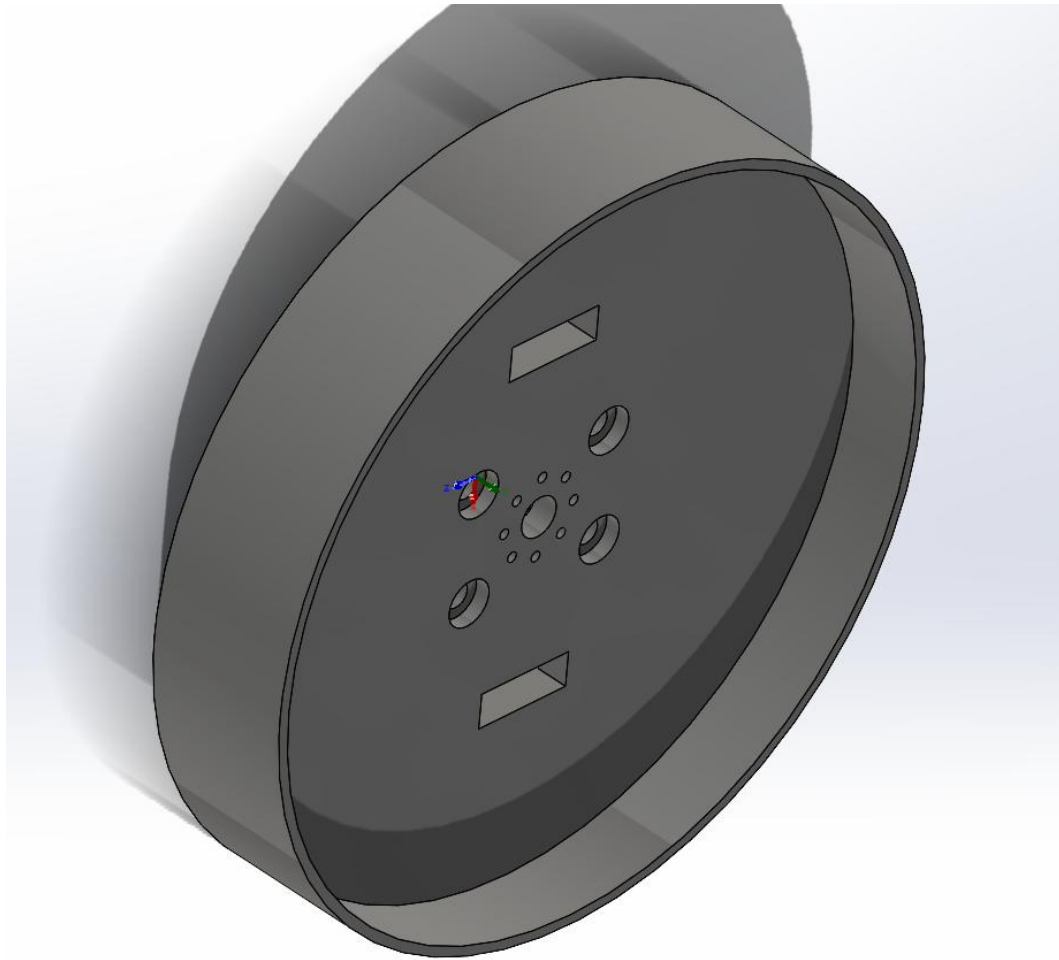


Figure 9.9 — Lid underside, showing the cable routing pockets.

The lid went through four iterations. The first was inherited from Sortify. The second tried to add stiffener flanges on the underside but didn't fix the deflection problem we were seeing under load. The third tried to integrate the shoulder bracket into the lid as a single print, but the layer orientation of the bracket portion (printed flat against the lid) meant the bracket cracked along the layers under shoulder load. The fourth and final iteration is flat with a separately-printed XM-540 shoulder bracket bolted on — see §9.13.4 below.

9.13.4 XM-540 shoulder bracket and the print-orientation story

The XM-540 shoulder bracket took five iterations to get right, and four of those five iterations were about print orientation rather than about geometry. This is one of the most important fabrication lessons from the project, so it's worth recording in detail.

The bracket carries the entire arm above the shoulder — it sits on top of the lid, with four M2 × 15 mm screws running down from the top into the XM-540 motor body underneath, plus four more M3 screws underneath the lid that fasten the XM-540 motor body to the platform. The load path goes from the long beam tip, through the long beam, into the shoulder horn, into the XM-540 motor body, through the bolts, into the bracket walls, and from the bracket walls into the lid. The bolt-bearing surface inside the bracket walls is where the highest stress concentration lives.

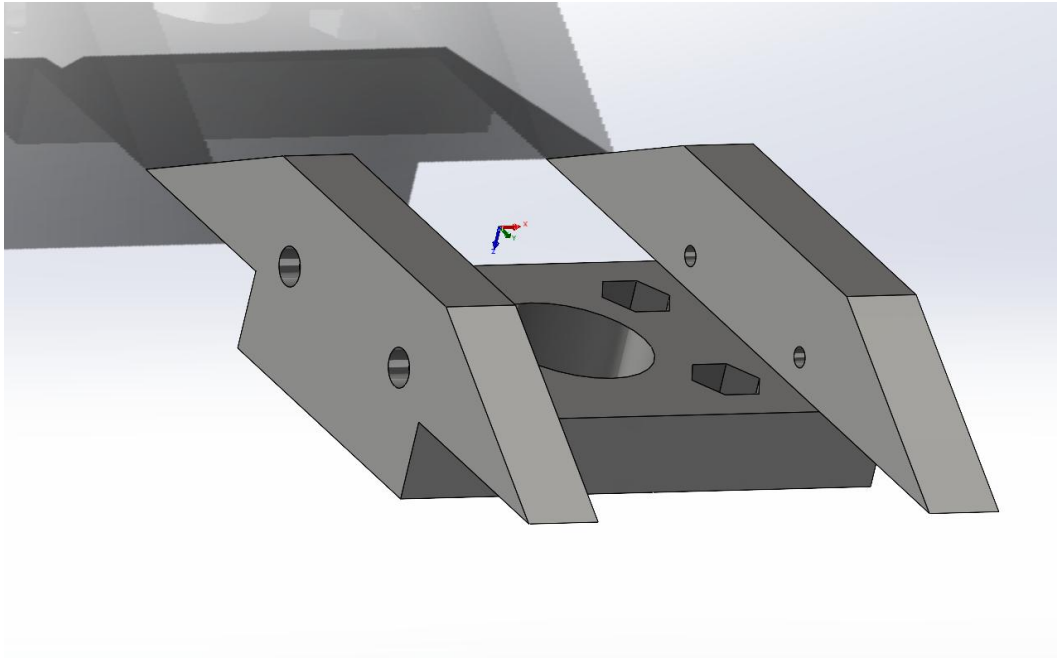


Figure 9.10 — XM-540 bracket iteration 1: printed with the motor cutout face up, supports on the top. The layer planes run perpendicular to the bolt-hole axis. Failed: cracked along the layers between the bolt holes under shoulder load.

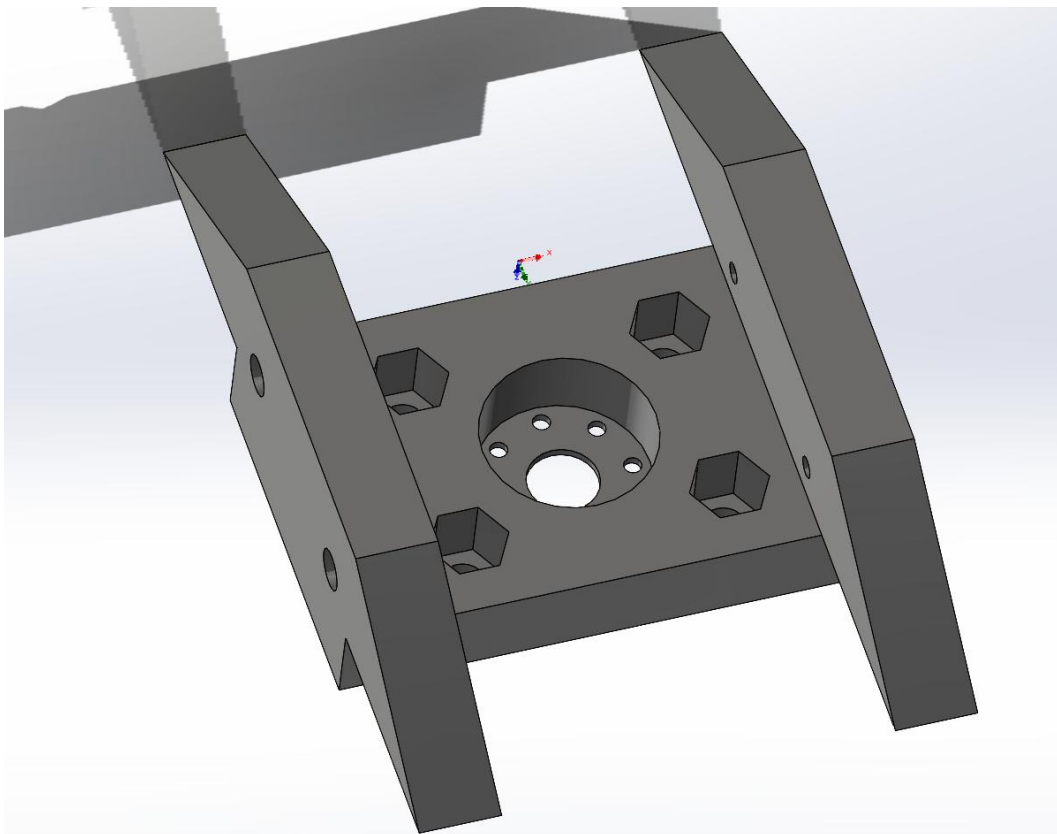


Figure 9.11 — XM-540 bracket iteration 3 with trapezoidal supports. Improved bolt-hole geometry but still printed face-up, so the layer-plane failure mode persisted.

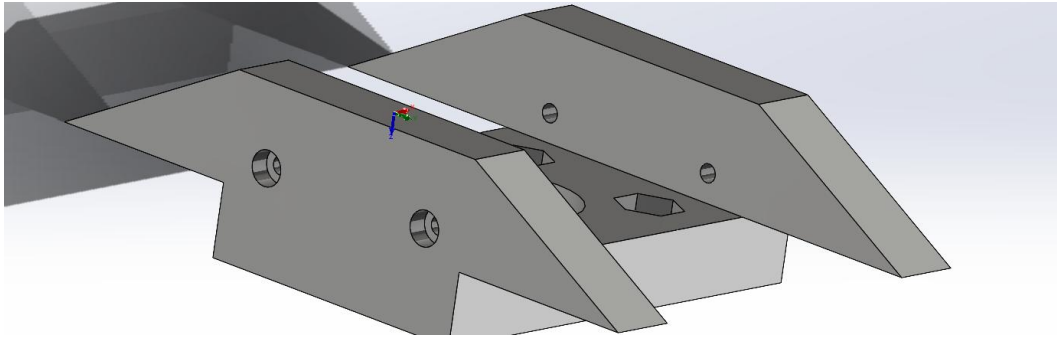


Figure 9.12 — XM-540 bracket iteration 5 (final): same geometry as iteration 4 but printed on its side, so the layer planes now run parallel to the load path through the bolt holes.

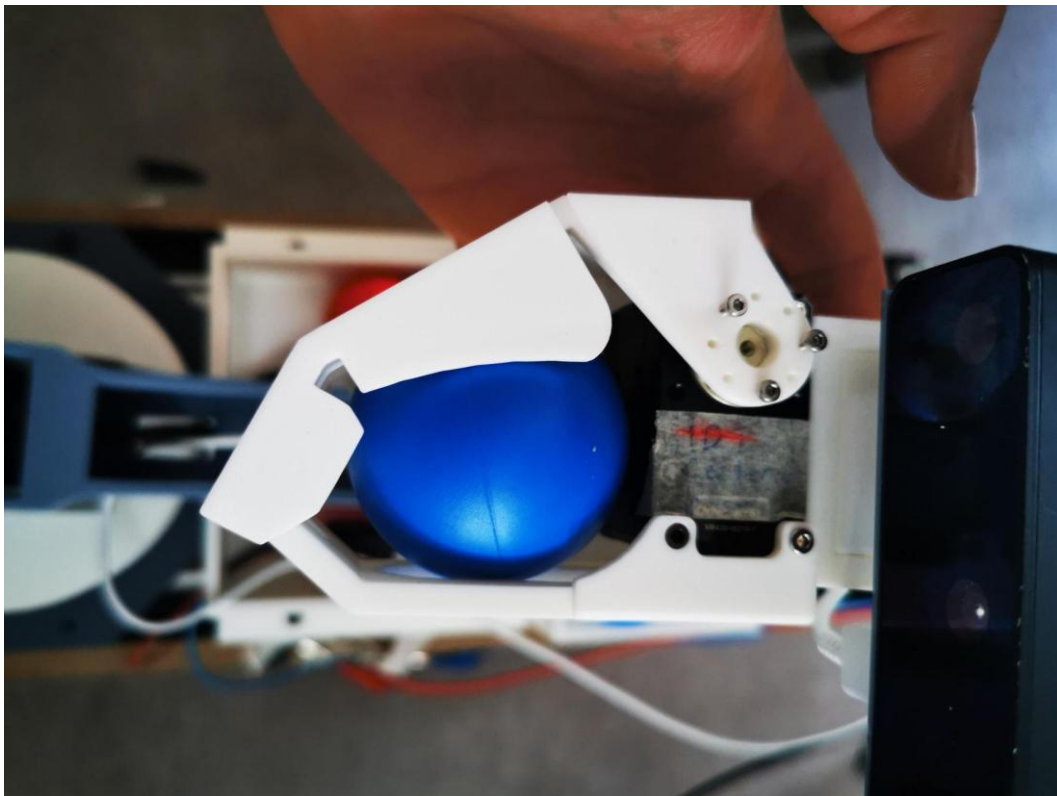


Figure 9.13 — An earlier-iteration XM-540 bracket showing the layer-plane crack between the two bolt holes. The crack runs across the full width of the part along a single horizontal layer, confirming the failure is layer-bond rather than bulk material failure.

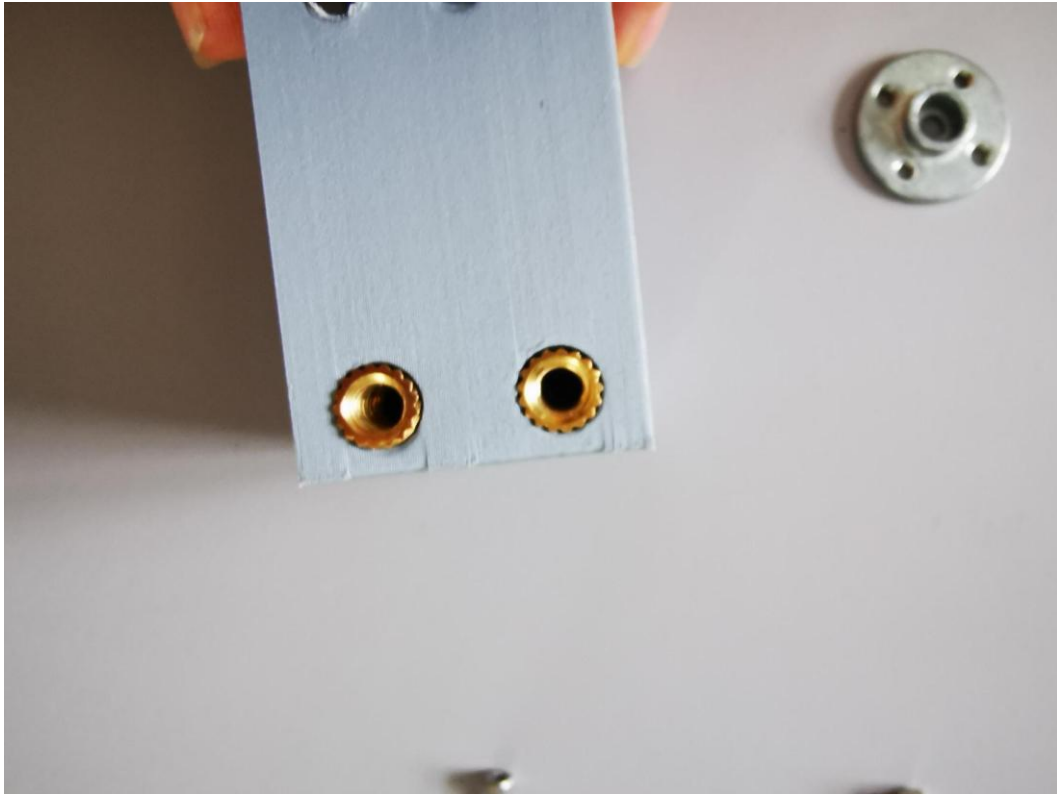


Figure 9.14 — Side-by-side comparison: failed early iteration (left) and the iteration-5 final design (right) with the integrated motor cutout, hex bolt pockets, and refined mounting plate geometry.

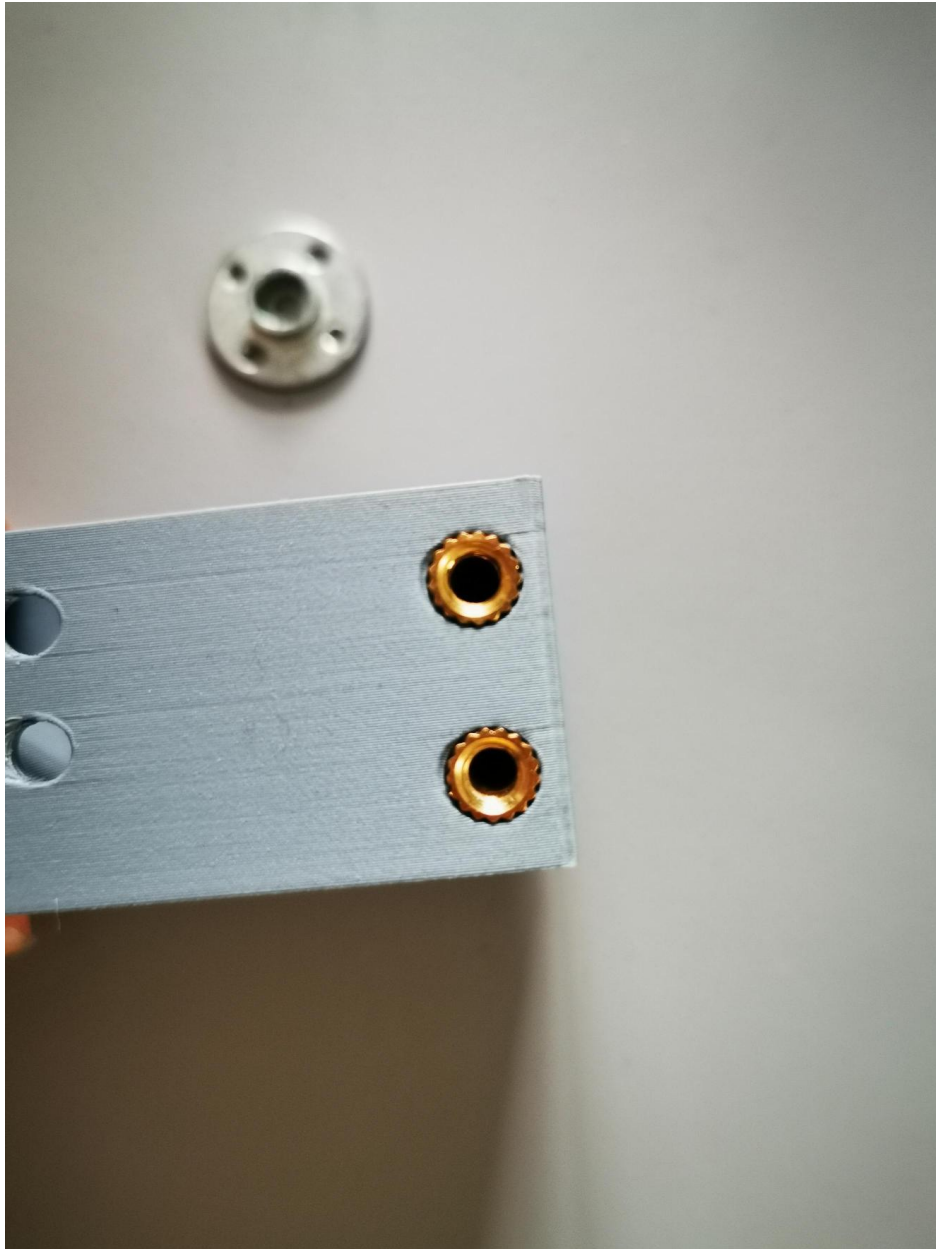


Figure 9.15 — Iteration-5 bracket exploded: the main bracket body (bottom, with the hex bolt pockets and motor cutout) and the separately-printed mounting plate (top, with the M3 bolt-pattern through-holes) that fastens the bracket assembly to the lid.

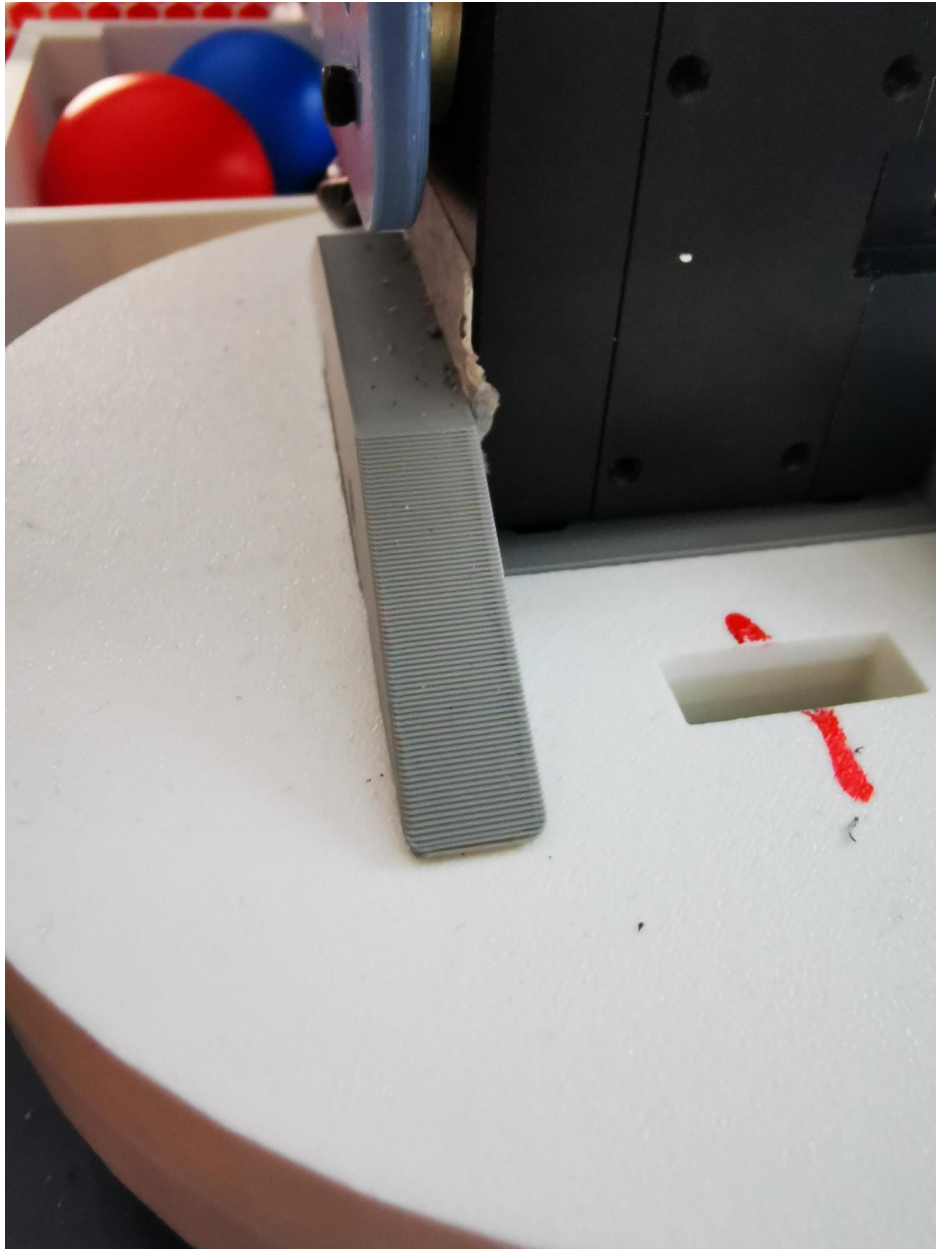


Figure 9.16 — Iteration-5 bracket installed in the lid, viewed from the side. M3 socket-head screws visible securing the bracket to the lid; the bracket leg has its layer lines running horizontally because of the sideways print orientation.

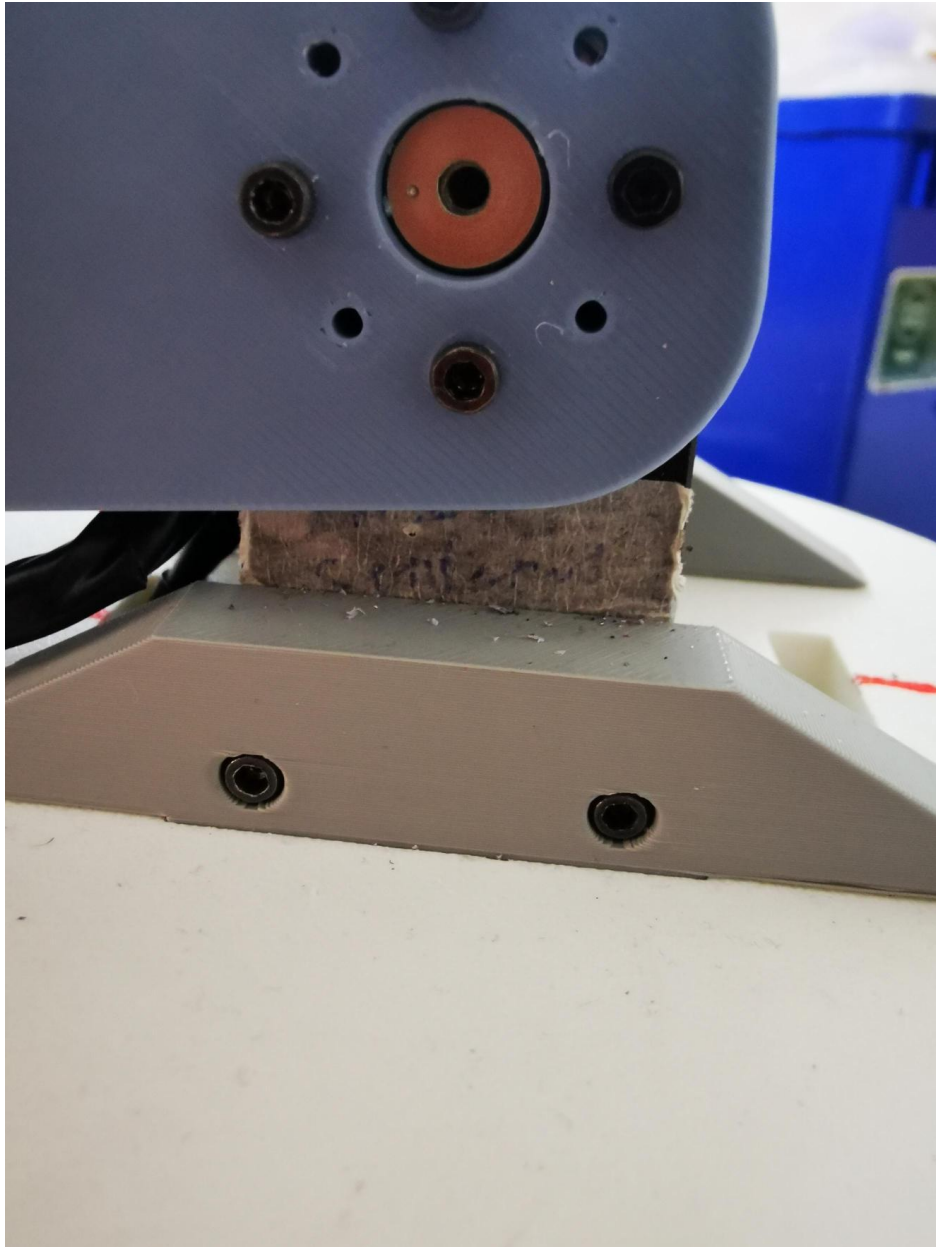


Figure 9.17 — Iteration-5 bracket from another angle, with the cardboard shim used to align the motor-to-bracket contact face during install (the shim was a temporary spacer during initial alignment and was removed for final fit).

The sequence of iterations was: iteration 1 printed face-up with no supports beyond the bed (failed at the bolt holes); iteration 2 printed face-up with brim and modified bolt pattern (failed in the same place); iteration 3 added trapezoidal external supports to the bolt-hole region (slightly better but still cracked); iteration 4 was the first attempt at a sideways print but with the bracket geometry still optimised for the original orientation — the part came off the printer with poor surface finish on what should have been the mounting face; iteration 5 is iteration 4 reworked so the mounting face is on the print bed and the bolt-hole axis is along the print build direction. The bolts now load the bracket walls along the layer direction (parallel to the layers) rather than across the layers, and the failure mode is gone.

The lesson: print orientation can matter more than geometry for layer-bond-limited features. If the load path runs across the layer planes, the part will fail no matter how thick you make the walls. Either reorient the print so the load runs parallel to the layers, or change the material to something effectively isotropic (Nylon 12 SLS, addressed in §9.18).

Base rotation assembly: platform with six bearings (washer-spaced) carrying the vertical load, rotation motor for torque only, lid bolted to the platform, XM-540 shoulder bracket bolted on top. The shoulder bracket's iteration history is the project's clearest example of print-orientation engineering: five iterations, four of them about printing direction.

9.14 Testing Against Requirements

Verification against the master mechanical requirements:

9.14.1 M1 — Workspace

The arm achieves 671.79 mm reach to the static-finger tip in the fully extended horizontal pose (§9.4.3), with 415.1 mm vertical drop to the gripper in the worst-case low-reach pose. Both numbers exceed the workspace required by the demonstration task. Verified at the system integration test stand.

9.14.2 M2 — Payload

The arm successfully grips and transports a 200 g ball, which is the project's target payload. The Factor of Safety analysis in §9.8 and §9.10 uses 200 g as the payload mass. The actual demonstration ran with balls in the 180–220 g range without any structural concerns observed; FEM and hand calc both confirm comfortable margin at the 200 g design point. Verified by inspection during integration testing.

9.14.3 M3 — Cycle time

Cycle time is set by the control system and the perception pipeline, not by the structural design. The arm has the actuator headroom to move faster than the demonstration cycle time requires (the joint motors are well below their rated continuous torque even at the worst-case static pose, so transient acceleration loads stay within bounds). System-level verification of M3 is in the integration chapter.

9.14.4 M4 — Durability

The arm completed all the demonstration runs without structural failure. The Factor of Safety values reported in §9.10 give substantial margin against single-event failure ($n = 2.53$ on the most-loaded section), and the print recipe was developed specifically to address the layer-bond failures that the early prints showed. No bracket cracking or beam deflection issues were observed during the demonstration runs. Verified by completion of the demonstration cycles.

All four mechanical requirements are met or exceeded. The arm reach, payload, cycle headroom, and durability targets are verified.

9.15 Final Dimensions and Totals

9.15.1 Mass breakdown

Mass of the as-built arm, broken down by subsystem:

- Long beam (CAD, 100 % infill): 219.38 g
- Middle beam (CAD, 100 % infill): 110.88 g
- Short beam (CAD, 100 % infill): 62.11 g
- End-effector printed parts (jaws, mounts, brackets): 91.5 g (sum of 14 + 25 + 13.52 + 14 + 5 + 5 + 14 + 1)
- Base rotation platform, lid, and XM-540 bracket (printed parts): ≈ 280 g
- Iteration-3 brackets and miscellaneous printed parts: ≈ 577 g
- Subtotal printed plastic (CAD, 100 % infill upper bound): 1341.22 g
- Motors (1 $\times$ XM-540 at 165 g + 4 $\times$ XM-430 at 82 g): 493 g
- Intel RealSense camera: 91 g
- Fasteners, inserts, cables, bearings (estimate): ≈ 100 g
- Total upper bound: 2025 g

The CAD plastic mass is the 100 % infill figure. At the print recipe in use (40–60 % gyroid infill, with shells/walls at the values listed in §9.6), the actual printed plastic mass is approximately 60–75 % of the CAD figure, so the realistic as-built total mass is approximately ~~1.49–1.55 kg~~[1.49–1.69 kg](#). The figure used in the cascade analysis (§9.5) is the CAD-mass upper bound, which is conservative for the structural calculations.

9.15.2 Overall dimensions

Overall arm dimensions in the fully extended horizontal pose:

- Reach from shoulder centre to short-beam tip: 601.76 mm
- Reach from shoulder centre to claw tip: 671.79 mm
- Vertical stack from rotation platform to XM-540 mounting face: 58.95 mm
- Worst-case low-reach vertical drop (shoulder to gripper tip): 415.1 mm
- Worst-case low-reach straight-line distance (shoulder to gripper): 544.95 mm

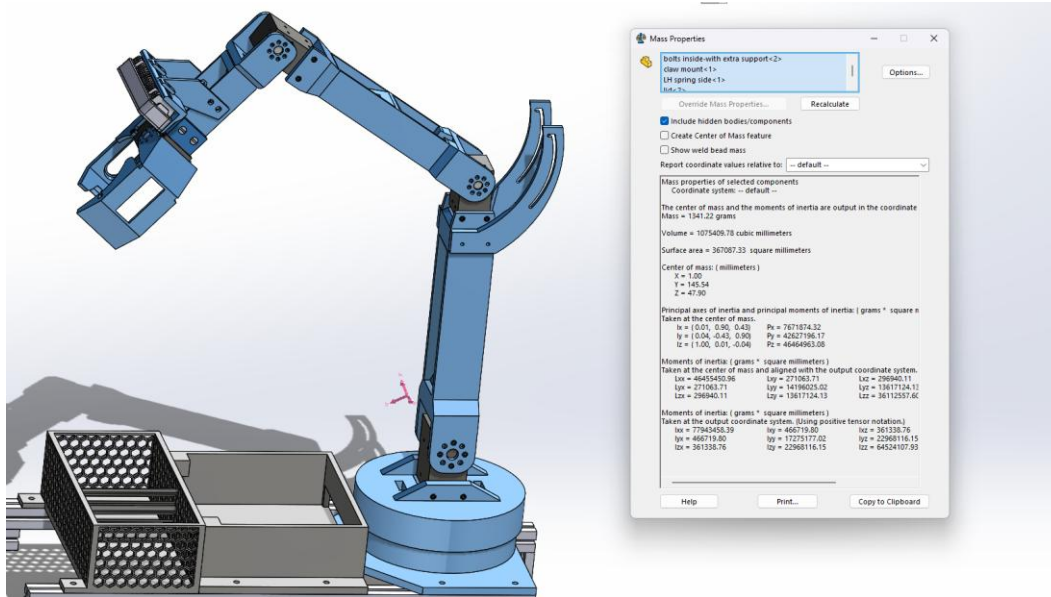


Figure 9.1 — Total CAD mass summary for the printed structural plastic parts, 1341.22 g at 100 % infill.



Figure 9.2 — The complete as-built arm in working configuration on its test rig, viewed from a different angle to show the base rotation platform, lid, and shoulder bracket assembly.

As-built arm: ~~1.49–1.55 kg~~1.49–1.69 kg realistic mass, 671.79 mm reach to claw tip, 5 active DoF. The arm is well within the structural strength margins required for the 200 g payload and meets the workspace requirement with margin to spare.

9.16 Limitations and Scoped-Out Analyses

We were honest with ourselves throughout the project about what we did and did not have time to do. The following items were scoped out of the mechanical design effort in this iteration, with a brief note on each:

9.16.1 Counterbalance bracket

A counterbalance bracket was designed and printed during the project but was not installed on the as-built arm. The design would have used a sprung mass on the rear of the long beam to offset the shoulder torque during static holds, reducing the average current draw on the XM-540 during long demonstration cycles. We chose not to install it because (a) the FoS and motor torque margins were comfortable without it ($n = 2.53$ on the long beam and shoulder torque well below continuous-rated), (b) installation would have required additional structural mods on the shoulder bracket that would have added schedule risk, and (c) the demonstration runs did not show any heat or current issues on the XM-540 that the counterbalance would have addressed. The counterbalance is therefore a designed-but-unused component; the printed parts exist and can be retrofitted in a future iteration if needed.

9.16.2 Mesh convergence study

No mesh convergence study was run on the FEM. Each beam was meshed once with a curvature-based mesh at a maximum element size between 7 and 11 mm depending on beam size, and the reported peak stresses are single-mesh estimates. Peak stress at a localised geometric stress concentration can shift by 5–15 % with mesh refinement, so a future-work convergence study would tighten the FoS numbers, particularly on the long beam where margin is tightest at $n = 2.53$.

9.16.3 LR 10K bending tests on iteration-3 beams

The compression tests in §9.11 were run on iteration-2 beam shells. The iteration-3 (as-built) integrated beams were not bending-tested on the LR 10K — the test fixture for cantilever loading was not available within the project schedule, and the FEM coverage (§9.10) was judged sufficient for the project's verification target. Physical bending tests on the as-built beams are recommended as future work.

9.16.4 Nylon 12 FEM completeness

Only the middle beam was modelled in Nylon 12 in the preliminary alternative-material study. The long beam and short beam Nylon 12 studies were not completed because the SLS service-bureau turnaround did not fit the project schedule. The Nylon 12 middle-beam result is presented in §9.18 as a worked example for future work.

9.16.5 CAD-confirmed cross-section dimensions

Two cross-section dimensions used in the hand calc are assumed rather than CAD-measured: the middle-beam X-width (assumed 36 mm) and the short-beam Z-depth (assumed 30 mm). These should be reconciled against the as-built CAD model in a follow-on pass; the FEM uses the actual CAD geometry so the FEM numbers are not affected, but the hand-calc bulk stresses for those two beams should be re-checked once the actual dimensions are confirmed.

9.16.6 Full bolt-torque specification

Bolt torques were not specified or recorded during assembly — the standard practice was “snug, not stripped” on the PLA-threaded holes (now backed up by Ruthex inserts on the short beam) and

standard wrench torque on the metal-to-metal joints. A formal torque schedule with values per bolt size and engagement length would be a reasonable addition for any production version.

Scoped-out items are documented here so a future group has a clear pickup list. None of them invalidate the verified Factor of Safety values; they are refinements rather than gaps.

9.17 Improvements

If we were starting the mechanical design over with what we know now, the changes we would make are:

9.17.1 Print-orientation-first design

The XM-540 bracket lesson — print orientation matters more than geometry for layer-bond-limited features — should be applied at the beginning of the design rather than discovered through five iterations of failure. Every load-bearing printed part on the arm should have its print orientation chosen up front based on the load path, before the geometry is fully fixed. In practice that means: identify the highest-stress feature on the part, identify the load direction, and orient the part so the load runs parallel to the layers, not across them. This costs nothing in print time and eliminates the layer-bond failure mode that took us five iterations to defeat on the shoulder bracket.

9.17.2 Brackets-and-beams separately, iterate fast, integrate at the end

The iterate-cheaply-with-separate-brackets methodology (iteration 2) was the right approach for getting to a working design quickly. We could test bracket variants in hours rather than days, and the bolted joints — while structurally weaker than integrated geometry — let us swap parts without reprinting the beams. The mistake was running the LR 10K compression tests on iteration-2 shells without brackets attached, which gave us shell-buckling data that wasn't directly useful for the iteration-3 integrated design. A future version would test the iteration-2 brackets-on-beams configuration first to validate the bolted interface, then move to integrated brackets only once the bracket geometry was nailed down.

9.17.3 Counterbalance integration

The counterbalance bracket exists but isn't installed. A future iteration with a longer project timeline would install it and characterise the reduction in average current draw and joint heating. The bracket design is ready; only the installation effort is scoped out.

9.17.4 Material change

The strongest single change available is moving to Nylon 12 SLS for the structural beams. This is the topic of §9.18.

9.18 Topology Optimisation and Material Future Work

9.18.1 Nylon 12 as an alternative material

Nylon 12 printed via Selective Laser Sintering (Formlabs Fuse or equivalent) was identified during material selection as a promising alternative to FDM PLA for the structural beams. The engineering advantages of SLS Nylon 12 over FDM PLA are substantial:

- Isotropic strength: SLS parts are effectively isotropic because the powder bed bonds in all three directions during sintering. There is no Z-axis weakness, no layer-plane failure mode, and no print-orientation dependency. The XM-540 bracket failure mode that took five iterations to defeat in FDM PLA would not exist in SLS Nylon 12.
- Ductility and impact resistance: Nylon 12 yields plastically before failing, which gives a built-in margin against overload events. PLA, by contrast, is brittle and snaps abruptly. For a manipulator subject to occasional collisions or unexpected loads, the ductile-yield-before-failure behaviour of Nylon 12 is a significant safety net.
- Fatigue resistance: Nylon 12 absorbs vibration and resists cyclic-load damage substantially better than PLA. For an arm that will see repeated motion cycles in service, this matters.
- Zero support structures: SLS uses unsintered powder as natural support, so the internal triangular ribs and any other complex internal geometry can be made without trapped-support issues. This opens up topology-optimised internal structures that are not practical in FDM.
- Low friction and wear: Nylon has natural low-friction behaviour, useful at pivot points and any sliding interface.

9.18.2 Preliminary Nylon 12 study

We ran a preliminary FEM study on the middle beam with the Nylon 12 material card ($\sigma_y = 50$ MPa, $E = 1.9$ GPa, $\rho = 1020$ kg/m³) using the same geometry and applied loads ($F = 3.745$ N, $T = 0.291$ N·m) as the PLA reference run [at the time. The 3.745 N tip load reflects an earlier version of the cascade before the gripper drive motor was finalised; the final cascade value for the middle beam is 5.82 N \(§9.5\). The PLA and Nylon 12 studies are therefore directly comparable to each other at 3.745 N, but neither reflects the final cascade load. Re-running both at 5.82 N is one of the items in §9.16 future work](#). Results: peak von Mises stress 4.33 MPa, peak displacement 0.627 mm, FoS ≈ 11.5 (against Nylon 12 yield). Compared to the PLA reference on the same beam (peak σ_{vM} 3.32 MPa, displacement 0.219 mm, FoS 10.5 against PLA yield), the Nylon 12 study confirms the expected trade-off: higher strength margin (because Nylon 12 has a higher yield stress) but more deflection (because Nylon 12 has a lower Young's modulus). The middle beam Nylon 12 case shows approximately 2.9× more deflection than the equivalent PLA case at the same load.

The trade-off generalises across all three beams: switching to Nylon 12 buys safety margin but costs stiffness. If we redesigned the arm for Nylon 12, we would need to thicken the walls or deepen the cross-sections to recover the stiffness, which would partially offset the mass savings from SLS's ability to use lighter geometries.

9.18.3 Why deferred

The reason we did not finish the Nylon 12 transition was the SLS service-bureau lead time. PLA prints can be iterated in hours on the in-house Bambu Lab printer; SLS parts have a typical service-bureau

turnaround of 5–10 working days plus shipping. With our project deadline, we could not afford a single round-trip to the service bureau, let alone the iteration cycle we would need to tune the cross-sections for Nylon 12's lower stiffness. The decision to stay with PLA was a schedule decision, not a strength decision.

9.18.4 Recommended follow-on work

If a future group continues the project with a longer schedule, the recommended Nylon 12 follow-on work is:

- Complete the Nylon 12 FEM on the long and short beams at full cascade loads, using the same methodology as the middle-beam preliminary study.
- Tune the cross-sections to maintain the PLA deflection target while keeping the Nylon 12 strength advantage — typically wall thickness + 1 mm or cross-section depth + 5 mm on each beam, depending on which beam dominates the deflection target.
- Manufacture the Nylon 12 beams via Formlabs Fuse 1+ (or equivalent SLS service bureau) and replace the PLA beams on the existing arm.
- Run comparison bending tests on the LR 10K (or equivalent) against the as-built PLA arm to confirm the predicted strength and deflection differences.
- Topology-optimize the internal rib structure using the freedom that SLS gives to use complex internal geometries that would be print-unsupportable in FDM.

9.18.5 Topology optimisation

The internal triangular rib structure used in the current beams is a hand-designed approximation of a load-path-optimised structure. With SLS Nylon 12 in hand, a future iteration could run topology optimisation (e.g. via the SolidWorks topology study or Altair Inspire) to generate organic load-path-following internal structures that minimise mass for a given stiffness target. This is the natural next step after switching materials — the material change opens up the design space and the optimisation tool exploits it.

Nylon 12 SLS is a recommended next step. The preliminary middle-beam study confirms the strength margin and quantifies the stiffness trade-off. Topology optimisation of the internal rib structure would be the natural follow-on once SLS manufacturing is in scope.

9.19 Chapter Conclusion

We set out to design and validate a 5-DoF robotic arm with sufficient reach, structural margin, and payload capacity to complete the project's pick-and-place demonstration task. We inherited the serial-revolute architecture and the Dynamixel motor set from the previous group (Sortify), redesigned every structural part in PLA on the in-house Bambu Lab P1S printer, and validated the redesign with hand calculation, FEM, and partial physical testing. The headline results:

- Reach: 671.79 mm to the claw tip in the fully extended horizontal pose, with a 415.1 mm vertical low-reach drop. Exceeds the M1 workspace requirement.

- Factor of Safety (FEM peak stress, full cascade load, $n = \sigma_y / \sigma_{vM,max}$): long beam $n = 2.53$, middle beam $n = 10.5$, short beam $n = 38.0$. All three beams exceed the $n \geq 2.0$ self-imposed target.
- Tip deflection at full cascade load: 0.25 mm long, 0.22 mm middle, 0.02 mm short. All deflections are visually imperceptible and well within positioning tolerance.
- Print recipe (40–60 % gyroid infill, 6–8 wall loops, 30–40 mm/s outer wall speed, 60 % max fan, 225–230 °C nozzle, Bambu PLA Basic on P1S 0.4 mm nozzle) developed to address the layer-bond failures seen in early iterations.
- Five iterations of the XM-540 shoulder bracket producing the key engineering finding of the project: print orientation matters more than geometry for layer-bond-limited features.
- Compression testing on iteration-2 beam shells (LR 10K) informed the move from separate to integrated brackets in iteration 3.
- Mass: ~~1.49–1.55 kg~~1.49–1.69 kg realistic as-built (≤ 2.03 kg upper bound at 100 % infill).

Scoped-out items that a future group should pick up: mesh convergence study, LR 10K bending tests on iteration-3 beams, full Nylon 12 FEM on long and short beams, counterbalance bracket installation, and CAD-confirmation of two cross-section dimensions used in the hand calc. None of these invalidate the verified Factor of Safety values — they are refinements rather than gaps.

The arm completed all demonstration runs in integration testing without structural failure. M1 through M4 are met or exceeded. The mechanical design supports the rest of the project.

Mechanical design verified: $n = 2.53$ on the most-loaded beam at full cascade load, 671.79 mm reach to the claw tip, 200 g payload, M1 through M4 satisfied. The arm is the foundation the rest of the project sits on, and that foundation holds.

Appendix A — Hand Calculation Worksheet

Full hand calculation, by beam. Repeats and extends the methodology summarised in §9.8. Equations and intermediate values are shown for traceability.

A.1 Shared inputs

- $\sigma_{yield} = 35$ MPa, $\sigma_{ultimate} = 30$ MPa, $E = 3.2$ GPa, $\nu = 0.394$
- Plane-stress von Mises: $\sigma_{vM} = \sqrt{(\sigma^2 + 3\tau^2)}$
- Bending stress: $\sigma = M \cdot c / I$
- Torsional shear (rectangular cross-section): $\tau_{max} = T / (\alpha \cdot b \cdot c^2)$
- Cantilever tip deflection: $\delta = F \cdot L^3 / (3 \cdot E \cdot I)$

A.2 Short beam

$L = 114.55$ mm, cross-section 36.42×30 mm, 2 mm walls. $I = 34,474$ mm⁴, $c = 15$ mm. $F_{tip} = 4.41$ N, $W_{beam} = 0.61$ N, $T = 0.069$ N·m.

$$R_y = 0.61 + 4.41 = 5.02 \text{ N}$$

$$M_{root} = 0.61 \times 57.3 + 4.41 \times 114.55 = 540.1 \text{ N·mm}$$

$$\begin{aligned}\sigma_z &= 540.1 \times 15 / 34,474 = 0.235 \text{ MPa} \\ \tau &= 69 / (0.262 \times 36.42 \times 225) = 0.0321 \text{ MPa} \\ \sigma_{vM} &= \sqrt{(0.235^2 + 3 \times 0.0321^2)} = 0.241 \text{ MPa} \\ n &= 35 / 0.241 = 145 \\ \delta &= 4.41 \times 114.55^3 / (3 \times 3200 \times 34,474) = 0.020 \text{ mm}\end{aligned}$$

A.3 Middle beam

L = 188 mm, cross-section 36 × 30 mm (assumed X-width), 2.5 mm walls. I = 40,635 mm⁴, c = 15 mm.
F_tip = 5.82 N, W_beam = 1.09 N, T = 0.291 N·m.

$$\begin{aligned}R_y &= 1.09 + 5.82 = 6.91 \text{ N} \\ M_{\text{root}} &= 1.09 \times 94 + 5.82 \times 188 = 1196.6 \text{ N·mm} \\ \sigma_z &= 1196.6 \times 15 / 40,635 = 0.442 \text{ MPa} \\ \tau &= 291 / (0.249 \times 36 \times 225) = 0.144 \text{ MPa} \\ \sigma_{vM} &= \sqrt{(0.442^2 + 3 \times 0.144^2)} = 0.508 \text{ MPa} \\ n &= 35 / 0.508 = 68.9 \\ \delta &= 5.82 \times 188^3 / (3 \times 3200 \times 40,635) = 0.099 \text{ mm}\end{aligned}$$

A.4 Long beam

L = 220.52 mm, cross-section 44 × 40.5 mm, 5 mm walls. I = 163,188 mm⁴, c = 20.25 mm. F_tip = 7.71 N, W_beam = 2.15 N, T = 0.889 N·m.

$$\begin{aligned}R_y &= 2.15 + 7.71 = 9.86 \text{ N} \\ M_{\text{root}} &= 2.15 \times 110.26 + 7.71 \times 220.52 = 1937.3 \text{ N·mm} \\ \sigma_z &= 1937.3 \times 20.25 / 163,188 = 0.240 \text{ MPa} \\ \tau &= 889 / (0.246 \times 44 \times 410.06) = 0.200 \text{ MPa} \\ \sigma_{vM} &= \sqrt{(0.240^2 + 3 \times 0.200^2)} = 0.421 \text{ MPa} \\ n &= 35 / 0.421 = 83.1 \\ \delta &= 7.71 \times 220.52^3 / (3 \times 3200 \times 163,188) = 0.053 \text{ mm}\end{aligned}$$

A.5 Bolt bearing

Worst-case bearing stress at the short beam camera mount: bolt force ≈ 1.7 N, bearing thickness ≈ 4 mm of PLA, M3 bolt diameter.

$$\sigma_{\text{bearing}} = 1.7 / (3 \times 4) = 0.14 \text{ MPa} \ll \text{allowable (typically 10–20 MPa for FDM PLA)}$$